DO RIDERS SPECIALIZE?

Clustering analysis on Tour de France riders, from **1994** to **2019**

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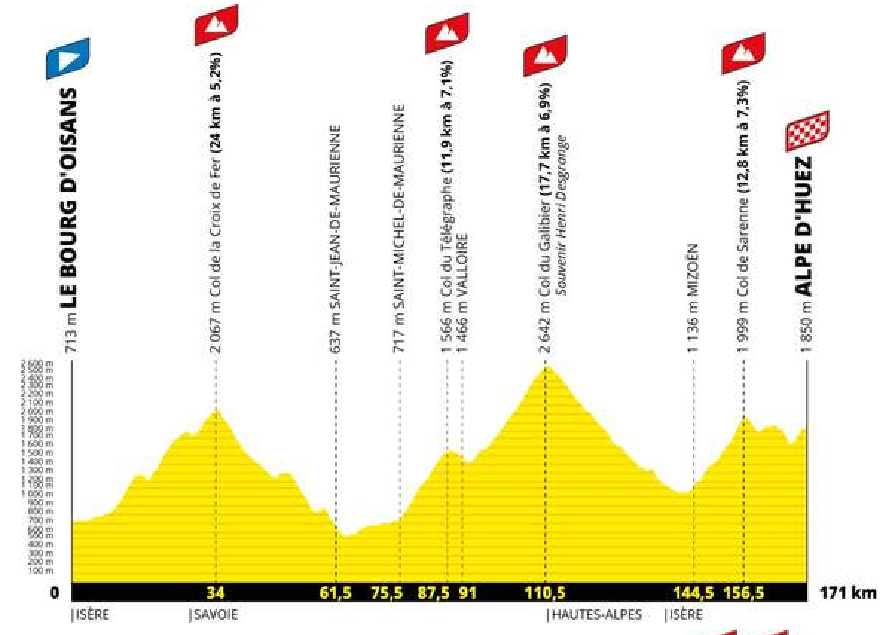
CONTEXT:

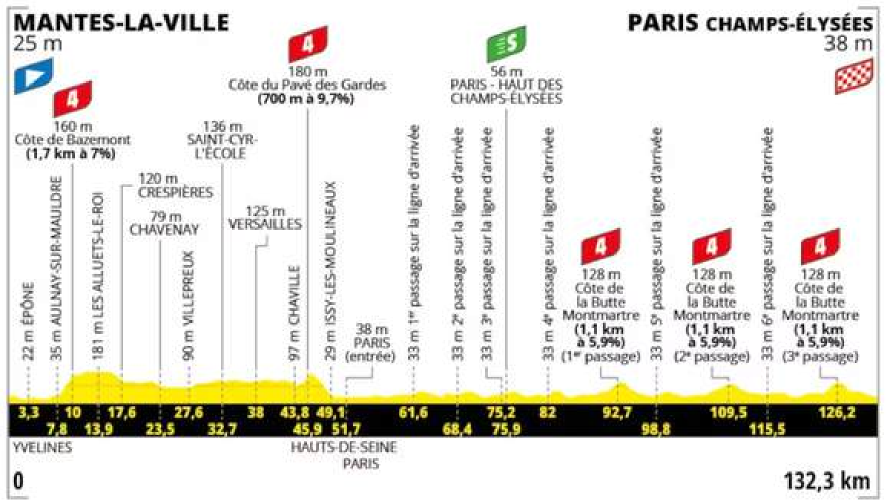
- The Tour de France is one of cycling's **three Grand Tours.**
- It is raced over **21 stages** (3 weeks)
- The peloton is composed of 22 professional teams, with 8 riders per team, for about **176 riders**
- Each teams combine **GC leaders, sprinters, stage hunters, domestiques, and young prospects**.
- Naturally **heterogeneous** dataset

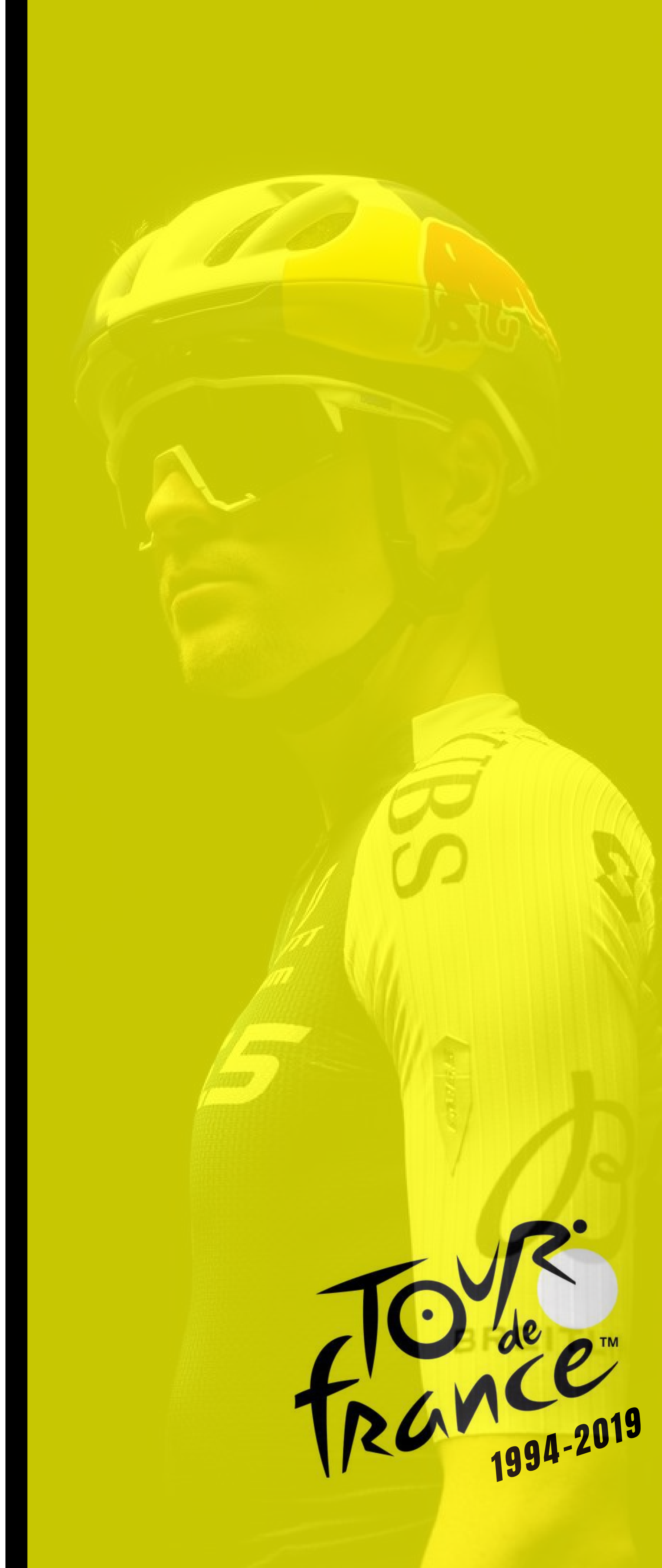
Tour de France
1994-2019

CONTEXT:

- Multiple stage types: **flat, hilly, mountain, and time trial stages.**
- 2026 route: **7** flat stages, **4** hilly stages, **8** mountain stages, **2** TT
- **Flat** stages often favor **sprinters.**
  Hilly stages reward **puncheurs** and breakaway specialists.
  Mountain stages and **TT** are usually more decisive for **GC riders**.







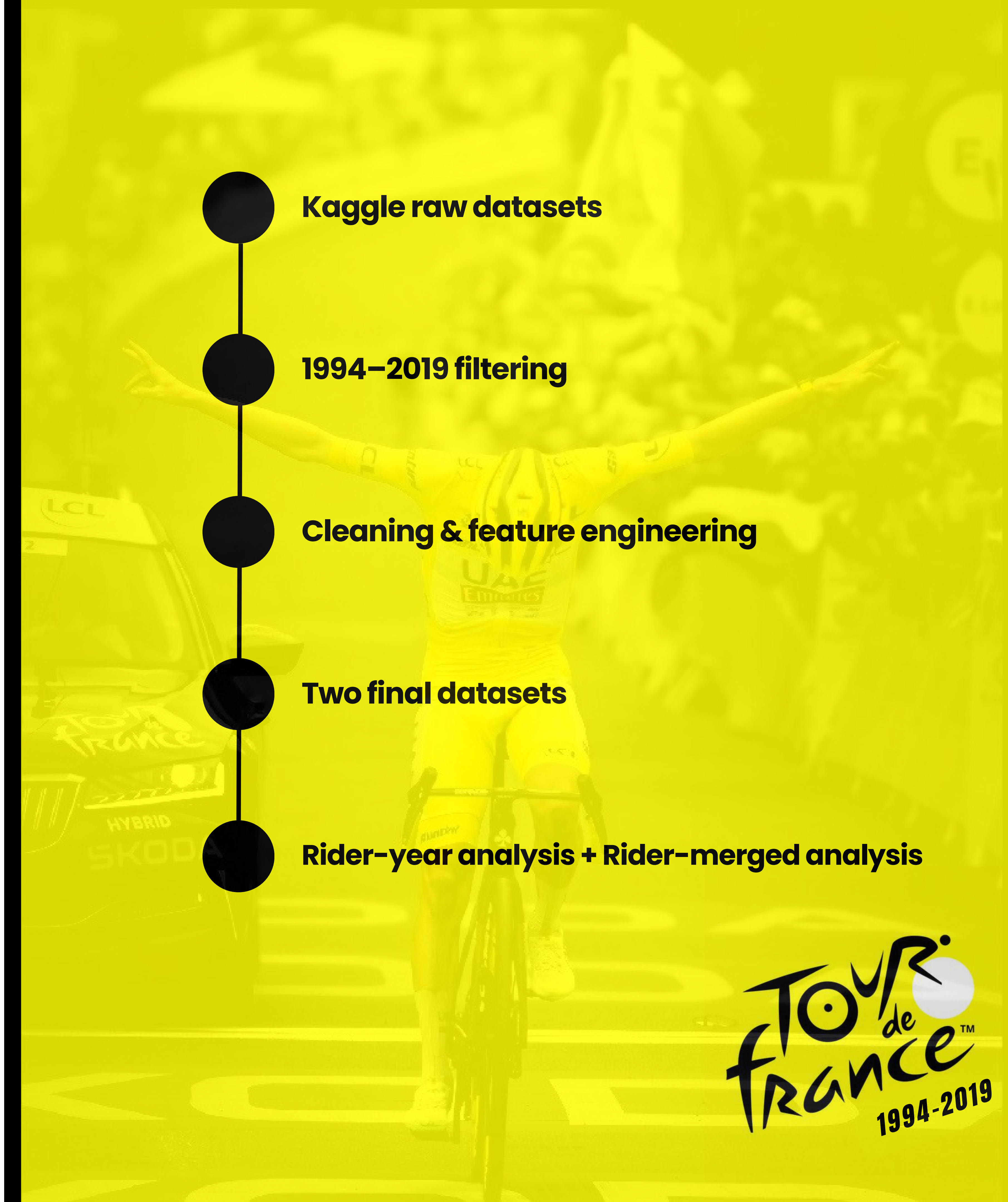
1994-2019

RESEARCH QUESTION:

“Can rider performances reveal meaningful specialization patterns?”

DATASET

- Historical Tour de France data from **Kaggle** *(stage results, finishers, and stage information)*
- We restricted the analysis to **1994–2019**
- We built two complementary datasets:
  Rider-per-year dataset *(One row = one rider in one specific Tour edition).*
  Rider-merged dataset *(One row = one rider aggregated across multiple Tour editions)*



CLEANING & FEATURES ENGINEERING:

TOUR de france 1994-2019

FEATURES ENGINEERING

- General performance
- Points for a specific stage type
- Top 10 finishes and wins for a specific stage type

NOTE: for the merged dataset

- Stage count features
- Percentage values and related metrics

NAN

- **Median** for age and percentage/related values
- **Zero** for others

OUTLIERS

- **IQR method and euclidean distance**
- Only **impossible** values were corrected.
- a log-transform was applied to point-based variables to reduce their impact in the rider-year dataset

SCALING

- Standard Scaling

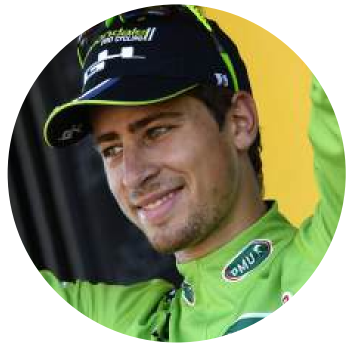
1 Analysis

RIDER PER YEAR

- Each observation represents a rider in a specific Tour edition.
- Each row corresponds to one rider in one Tour de France edition.
- The goal is to **identify annual performance profiles**, not fixed career identities.

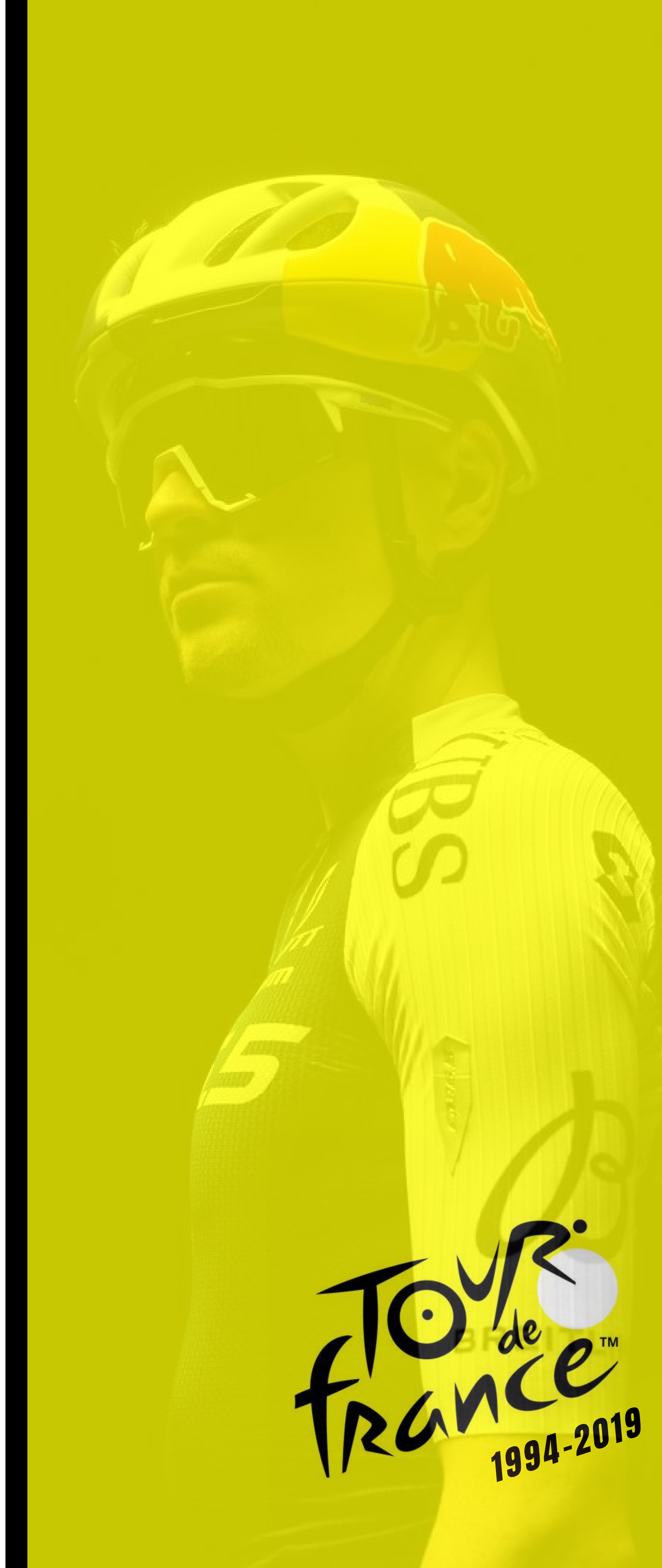
2012 2013 2014 2015








2016 2017 2018 2019

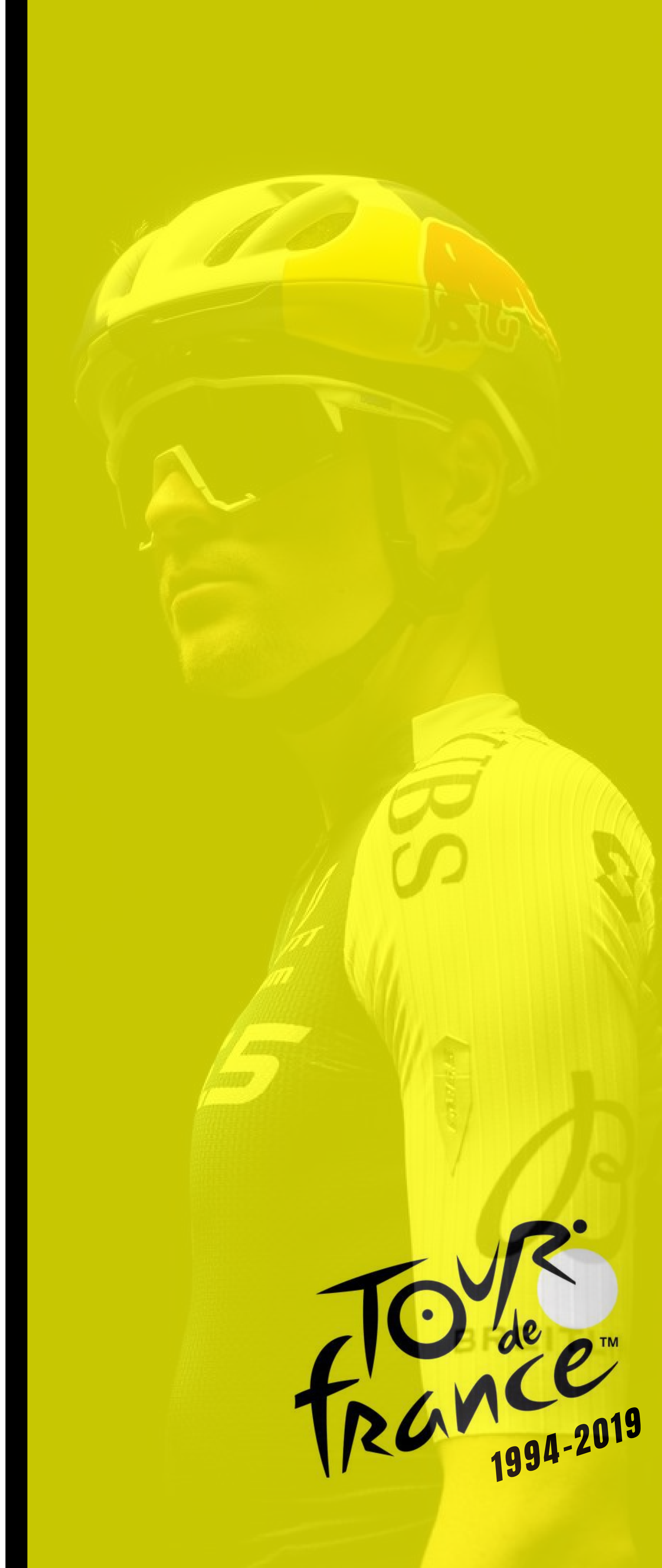


1 Analysis

PCA BEFORE CLUSTERING?

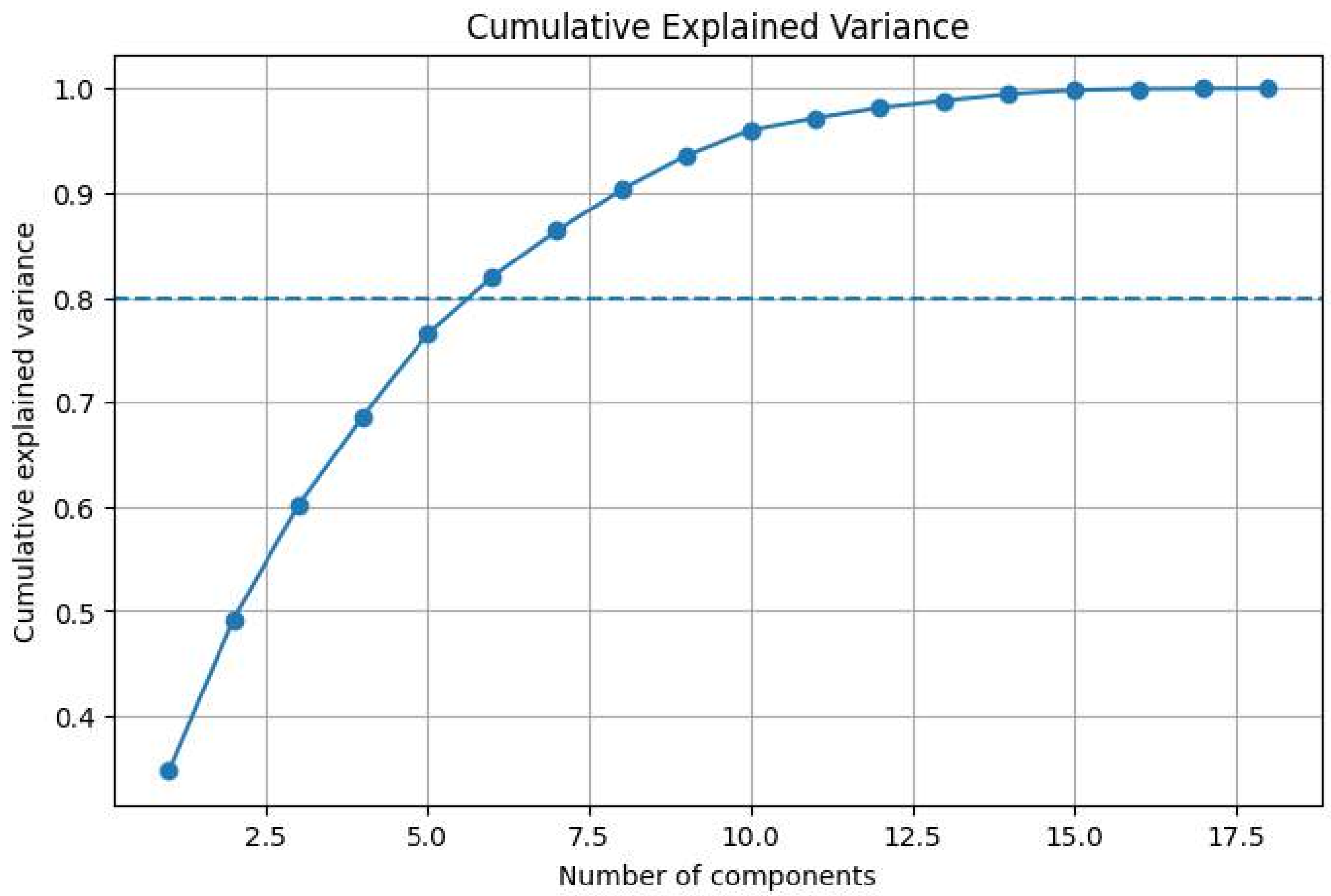
- We started from **18 standardized performance variables**, many of which are correlated

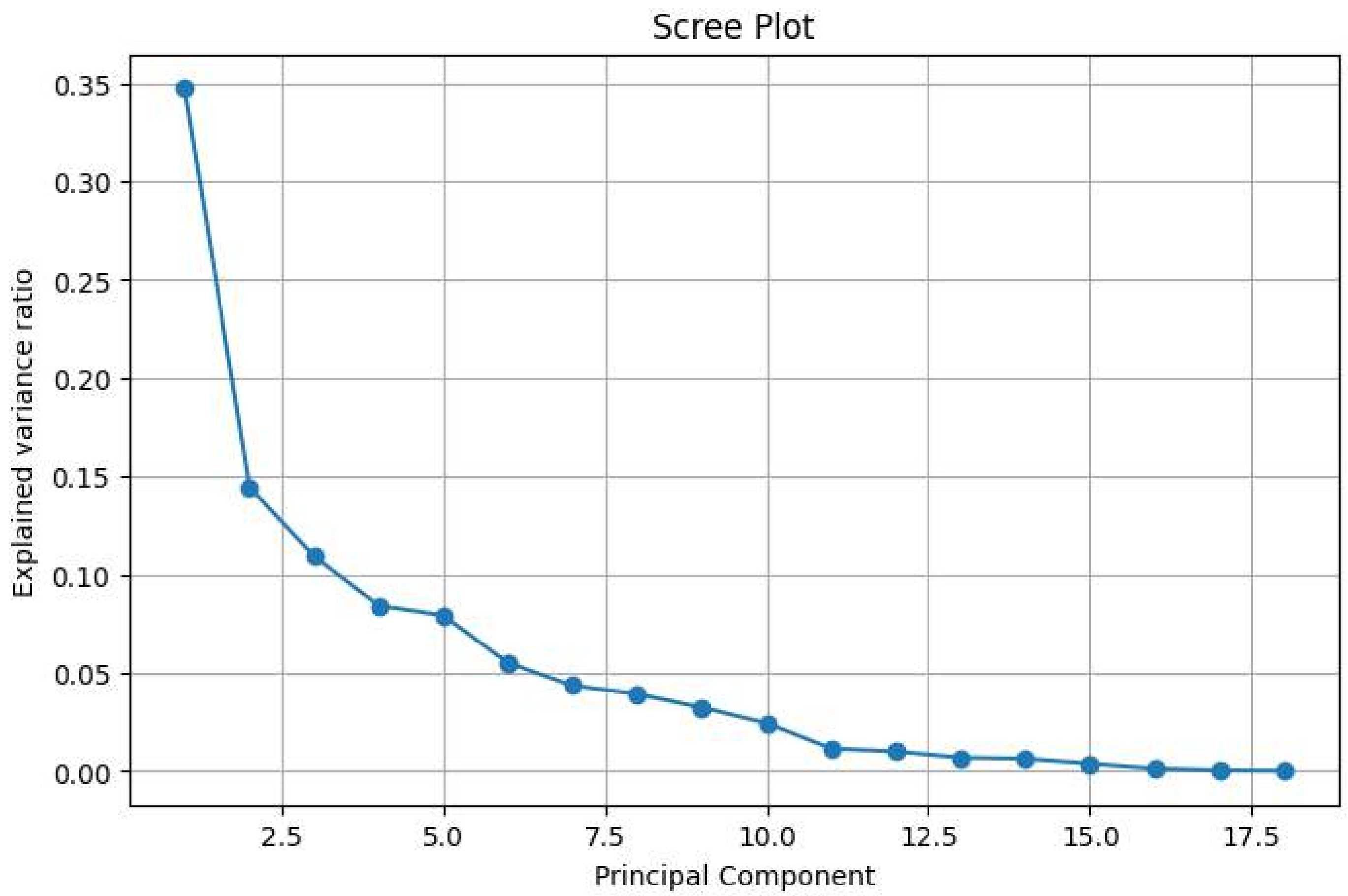
General Performance	stage_wins, top3_stages, top10_stages, log_total_points
Performance intensity	log_points_per_scheduled_stage, top10_per_scheduled_stage
Stage-Type top10:	top10_flat, top10_hilly, top10_mountain, top10_time_trial
Stage-Type Points:	log_points_flat, log_points_hilly, log_points_mountain, log_points_time_trial
Stage-Type wins:	wins_flat, wins_hilly, wins_mountain, wins_time_trial
Control	age

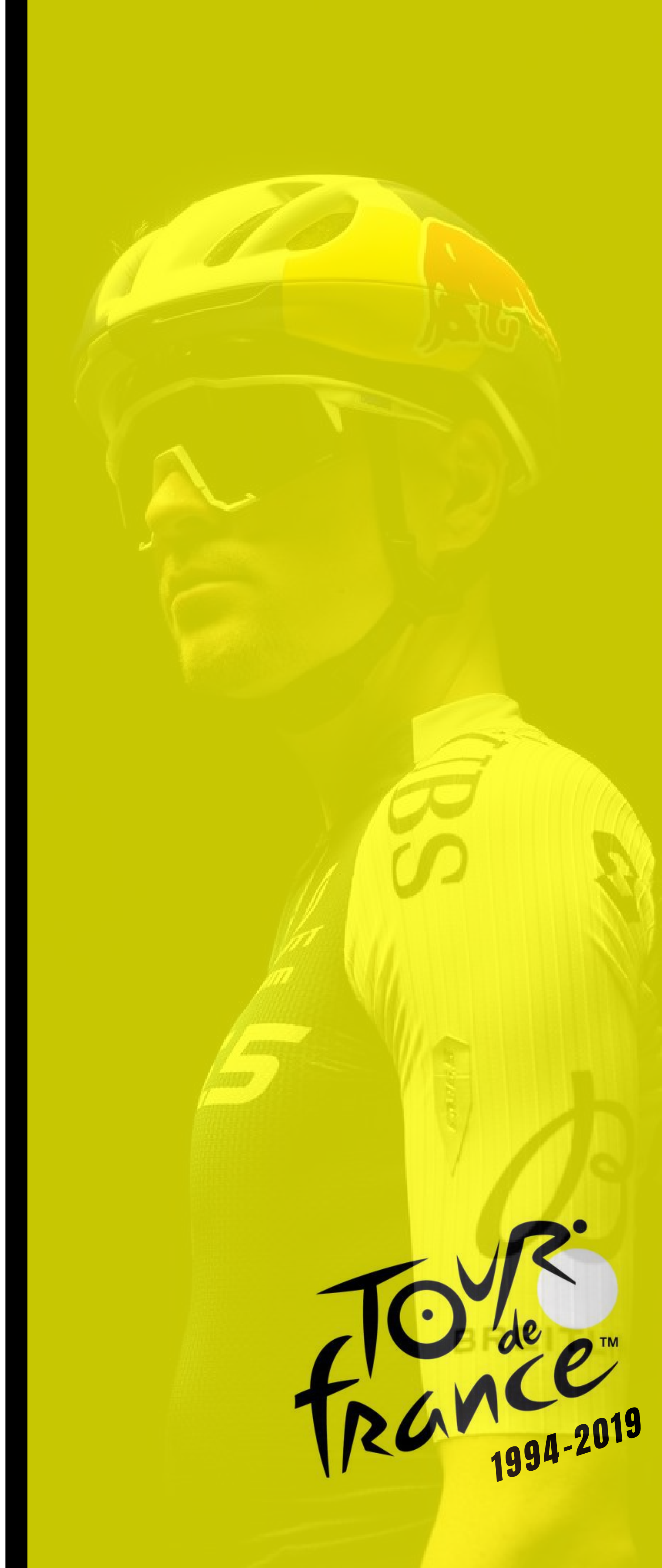


1 Analysis

PCA



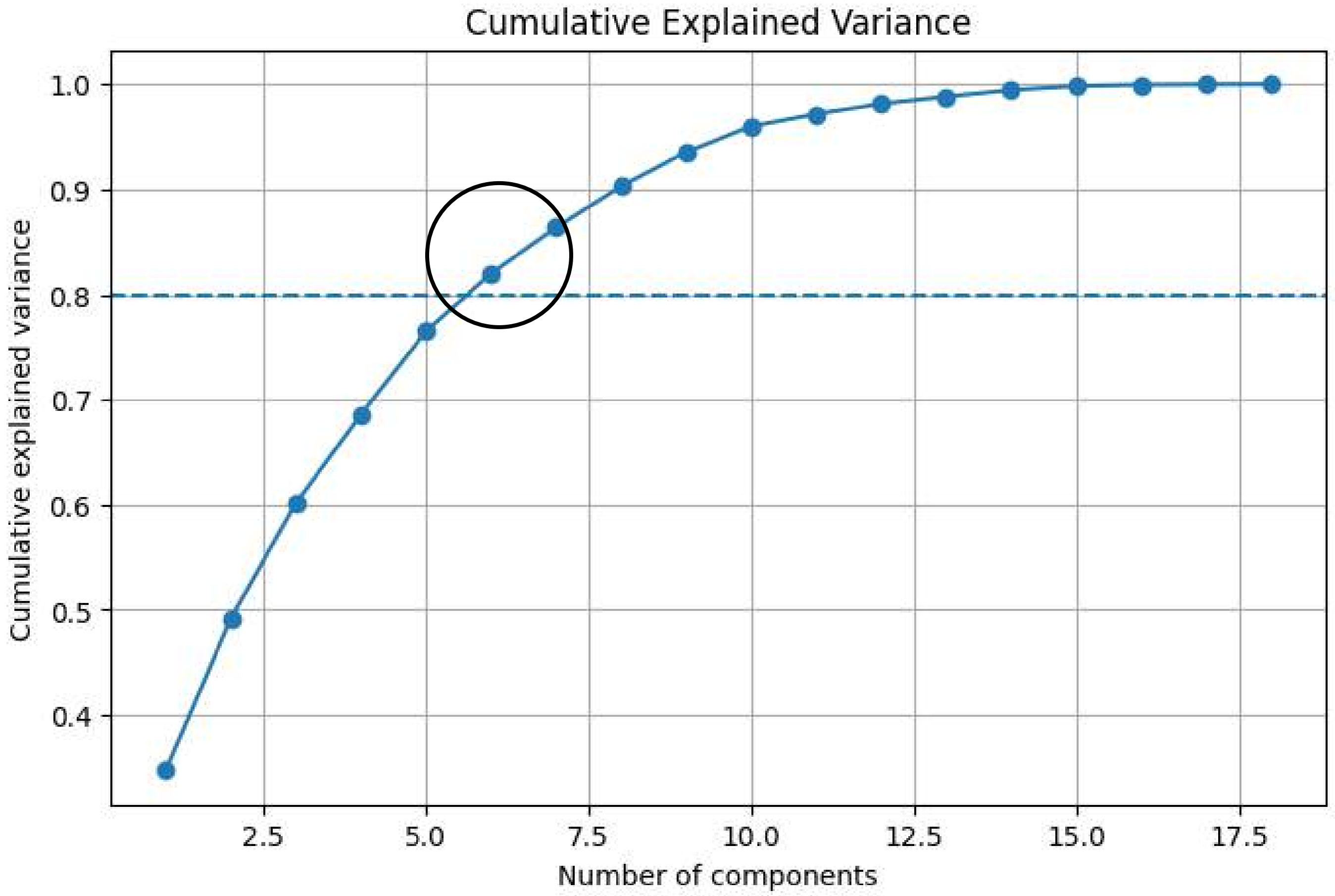


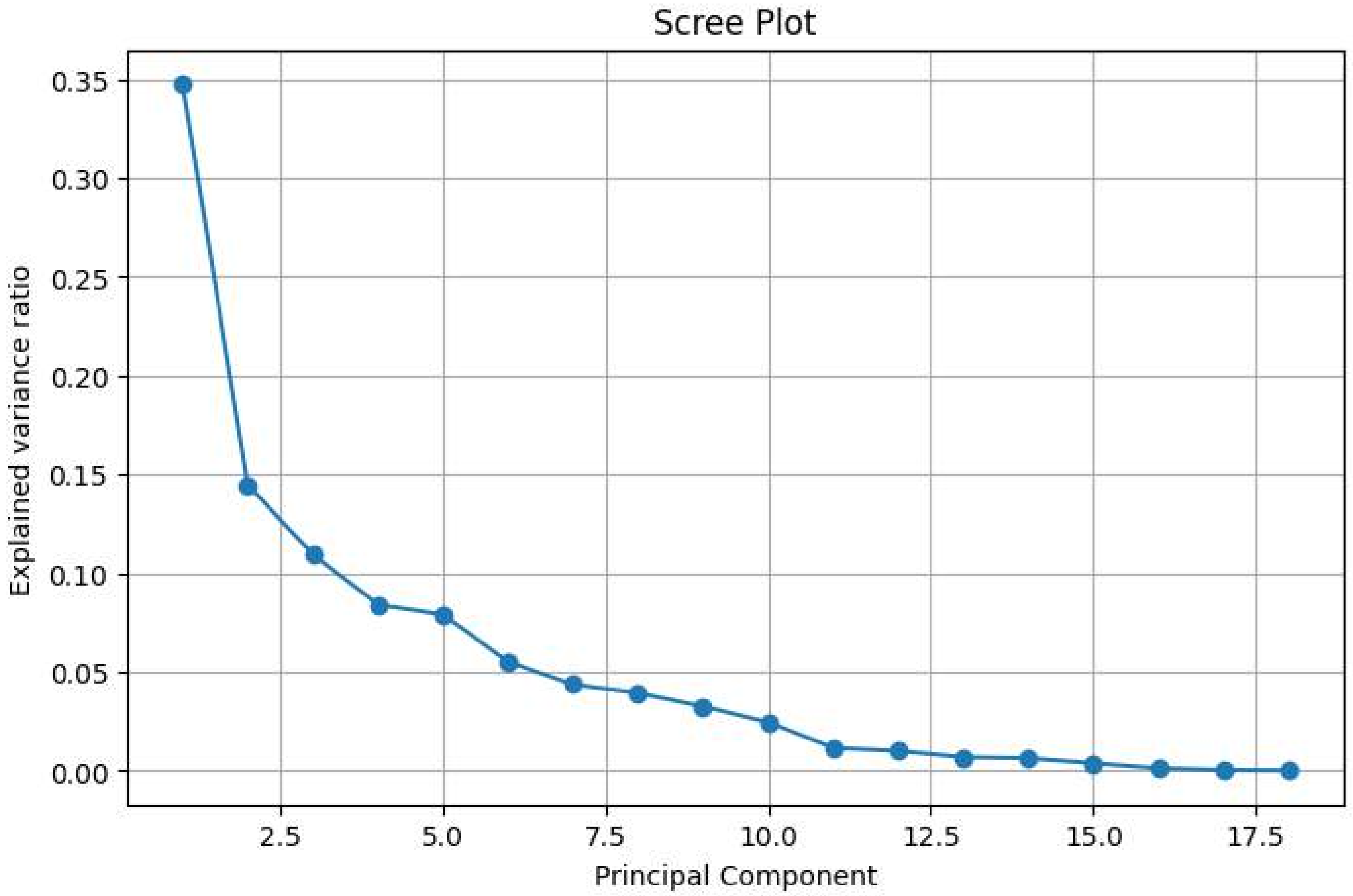


1 Analysis

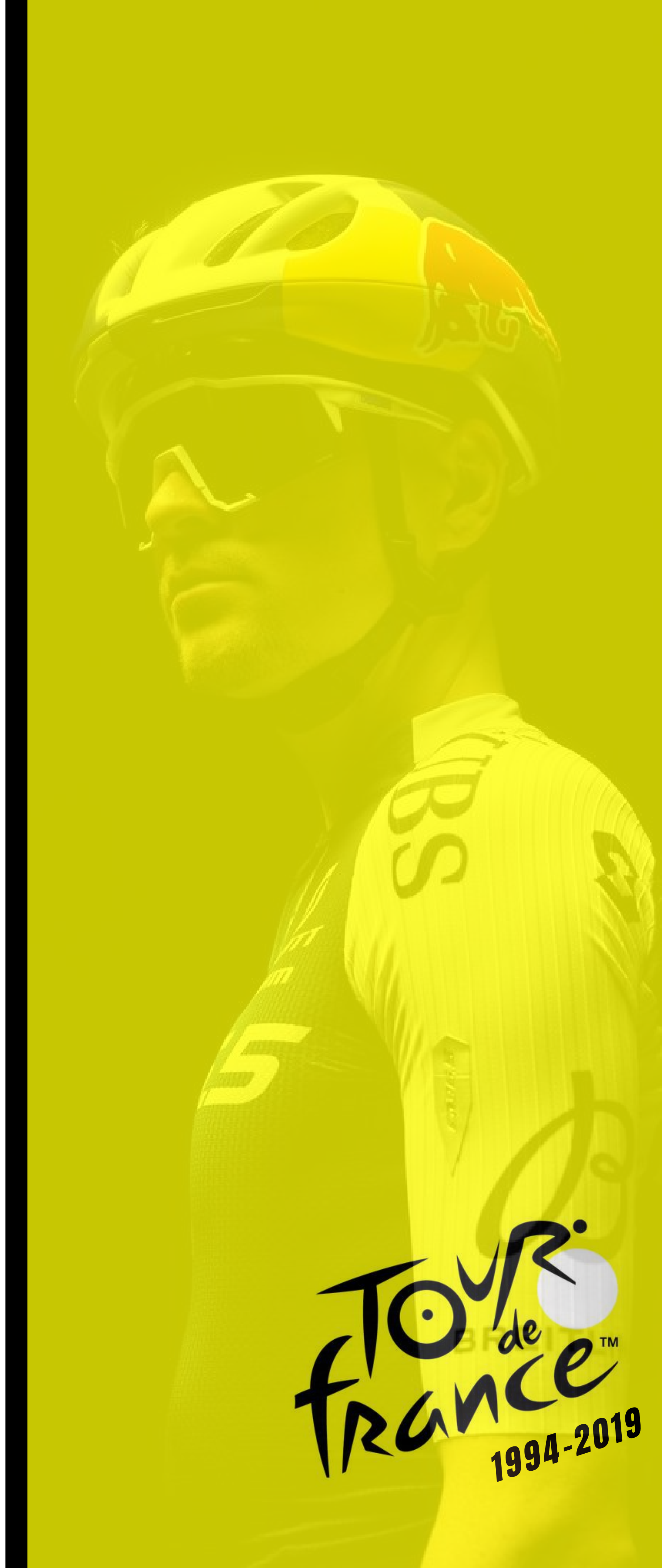
PCA

-The first 6 PC explain about **82.0% of the total variance**
-We used **the first 6 PCs for clustering,** keeping most of the information while reducing dimensionality.





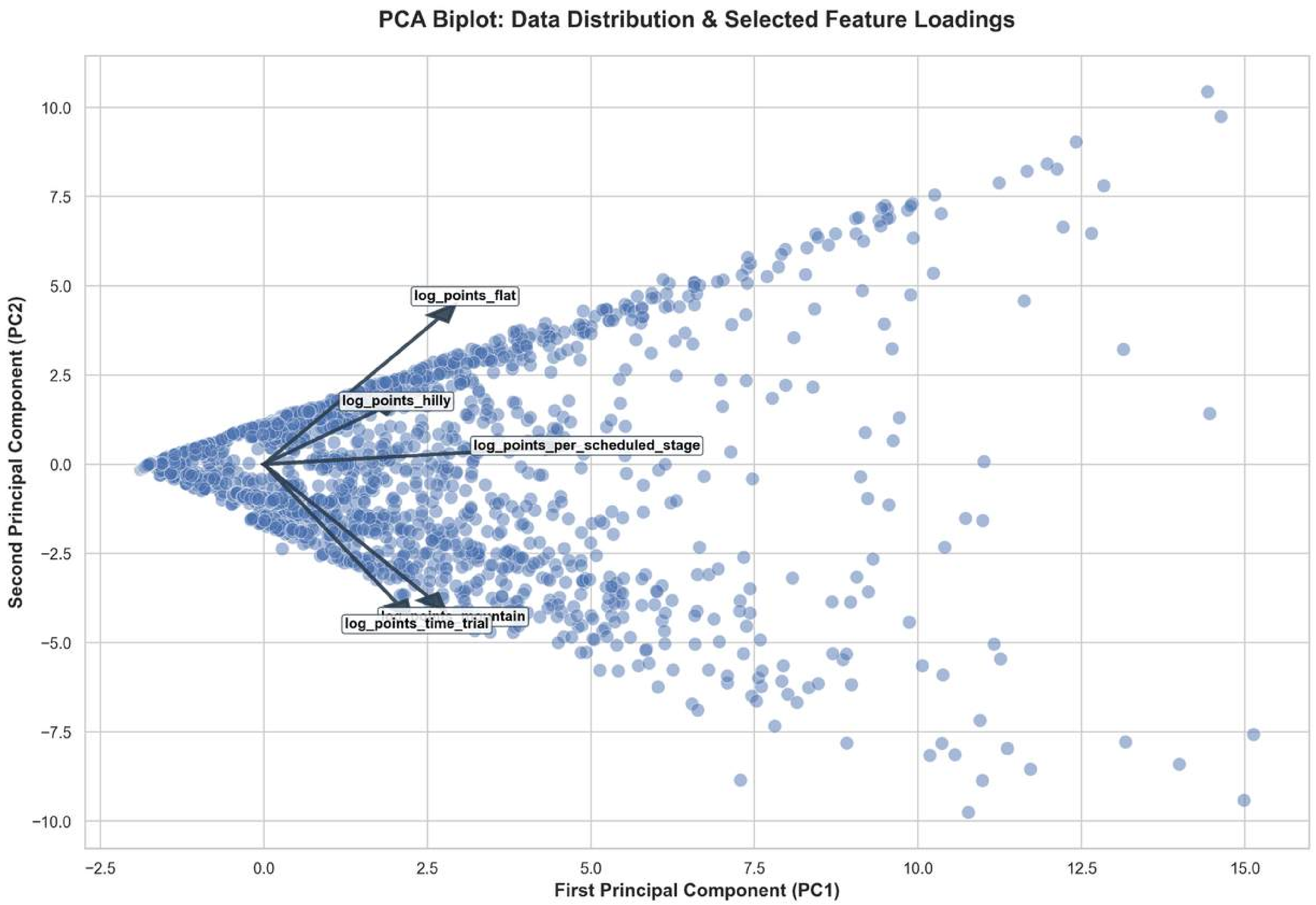
RETAINING 6 PRINCIPAL COMPONENTS



1 Analysis

BIPLOT: INTERPRETING PC1 AND PC2

- **PC1** (34.7%) and **PC2** (14.5%) explain **49.2%** of total Variance

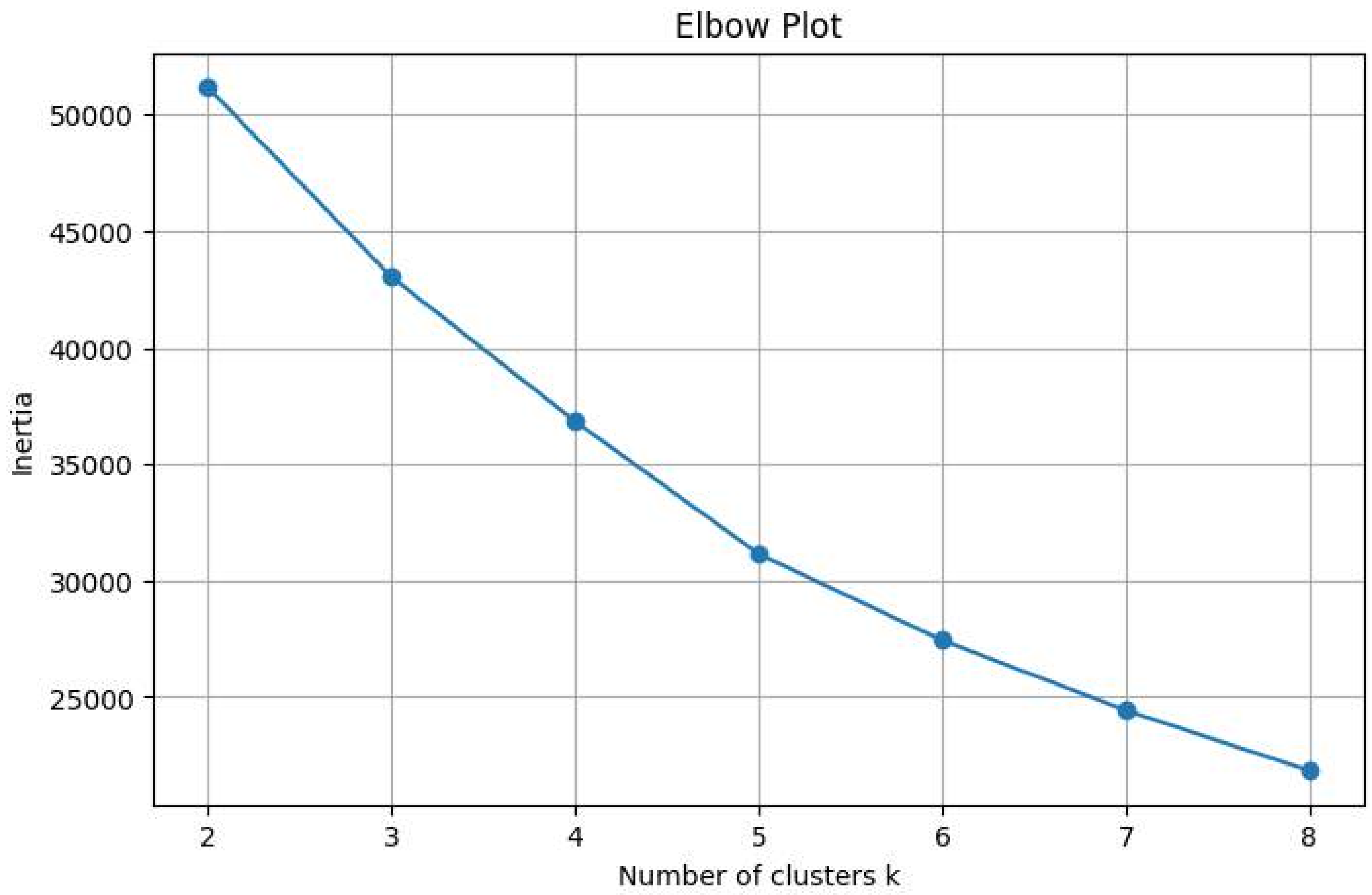


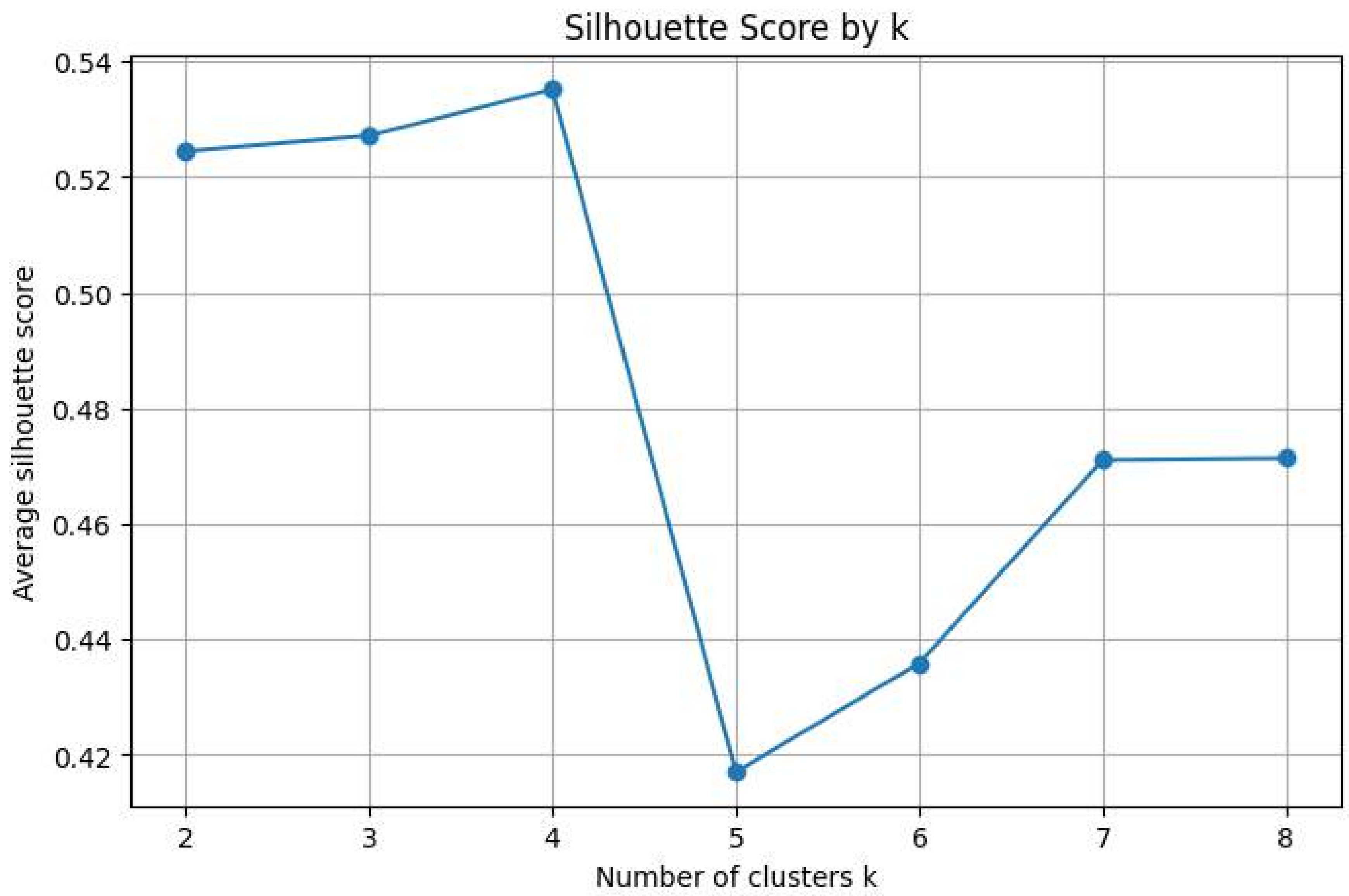
- **PC1** mainly represents **overall stage-performance intensity**.
- **PC2** separates **stage-type specialization**.
  (High: flat/hilly performance, negative: mountain and time-trial performance)

1 Analysis

K-MEANS ON **PCA SPACE**

- 18 standardized features → PCA → first 6 PCs → K-means clustering





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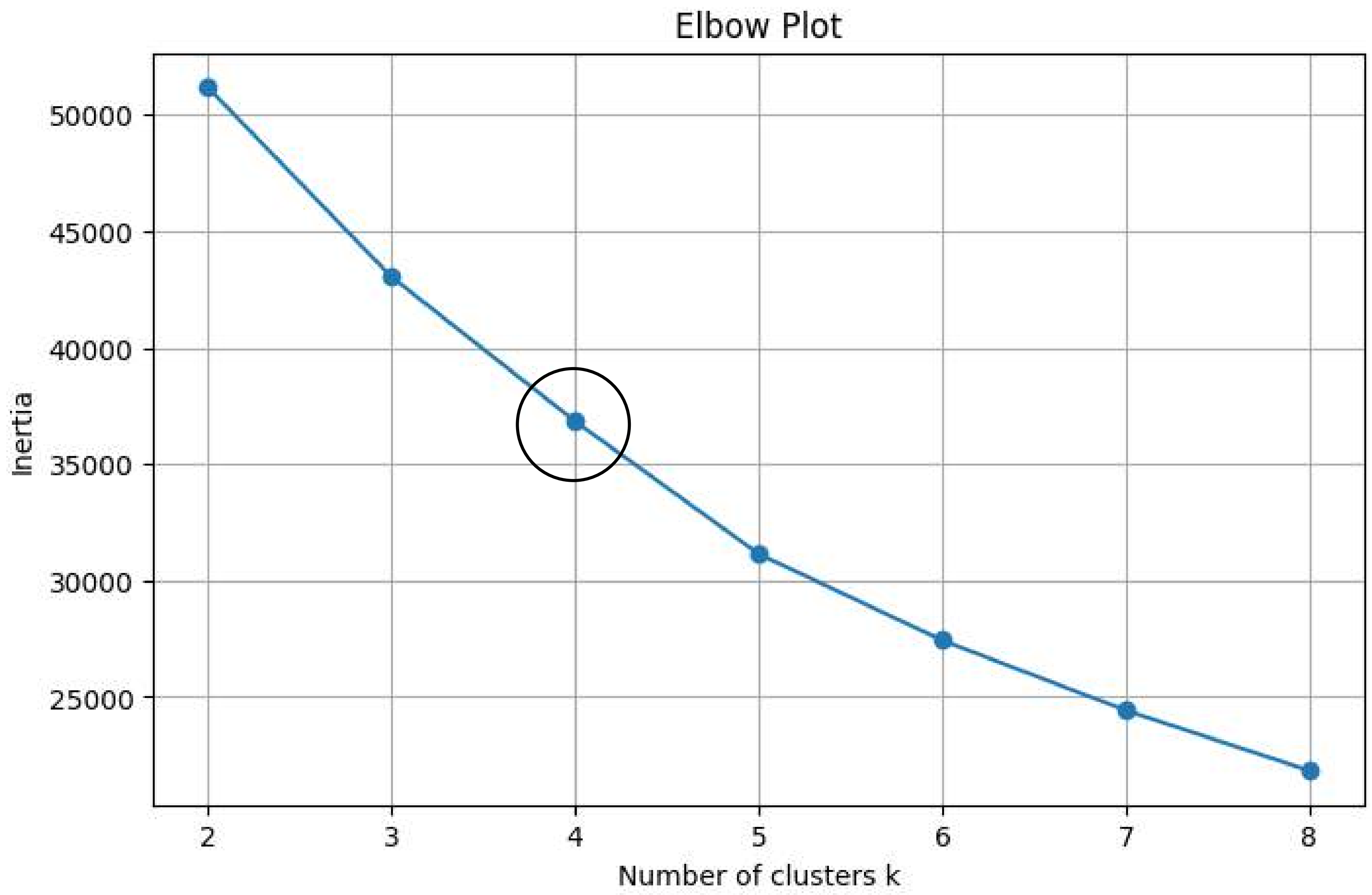
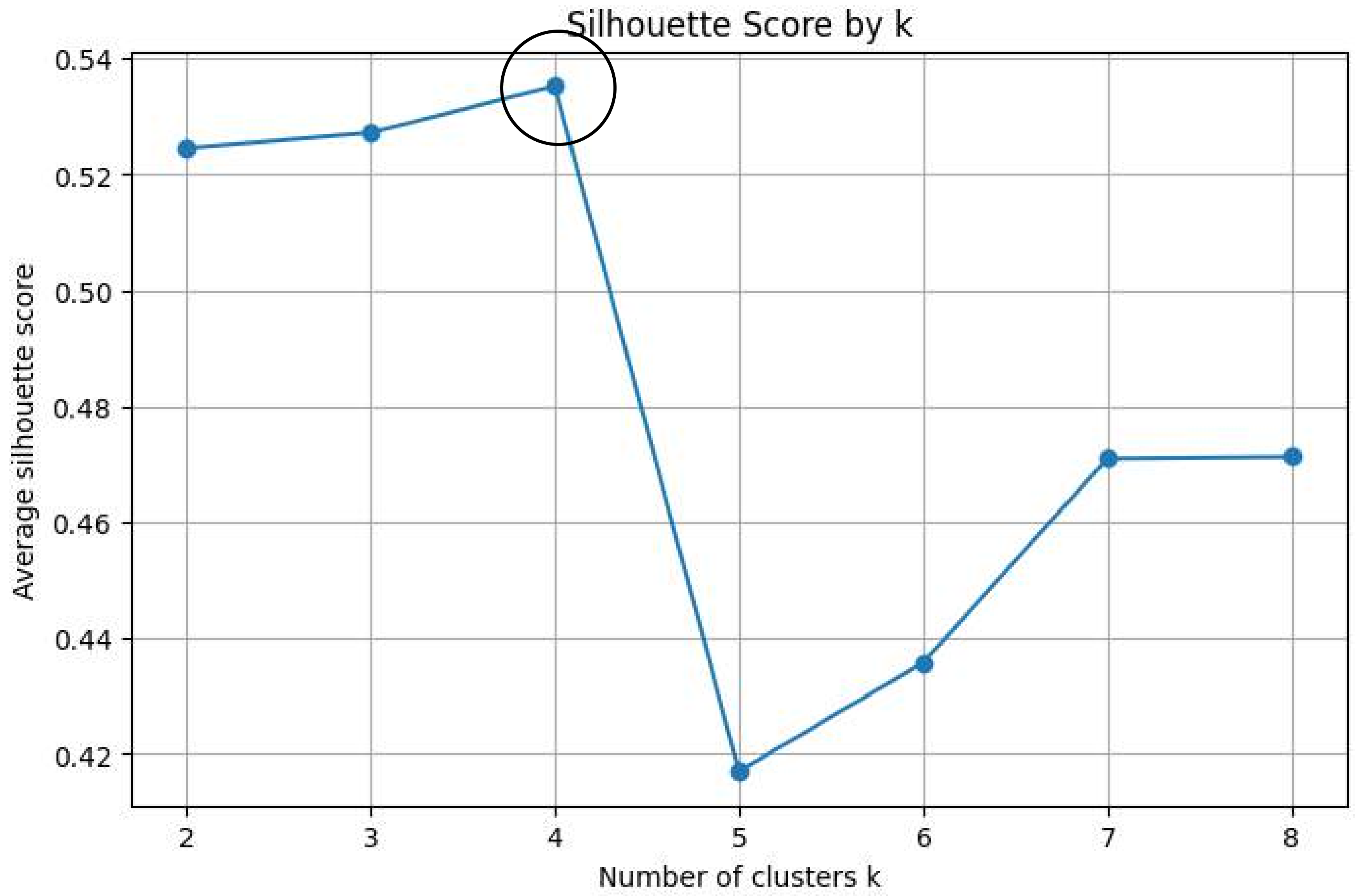
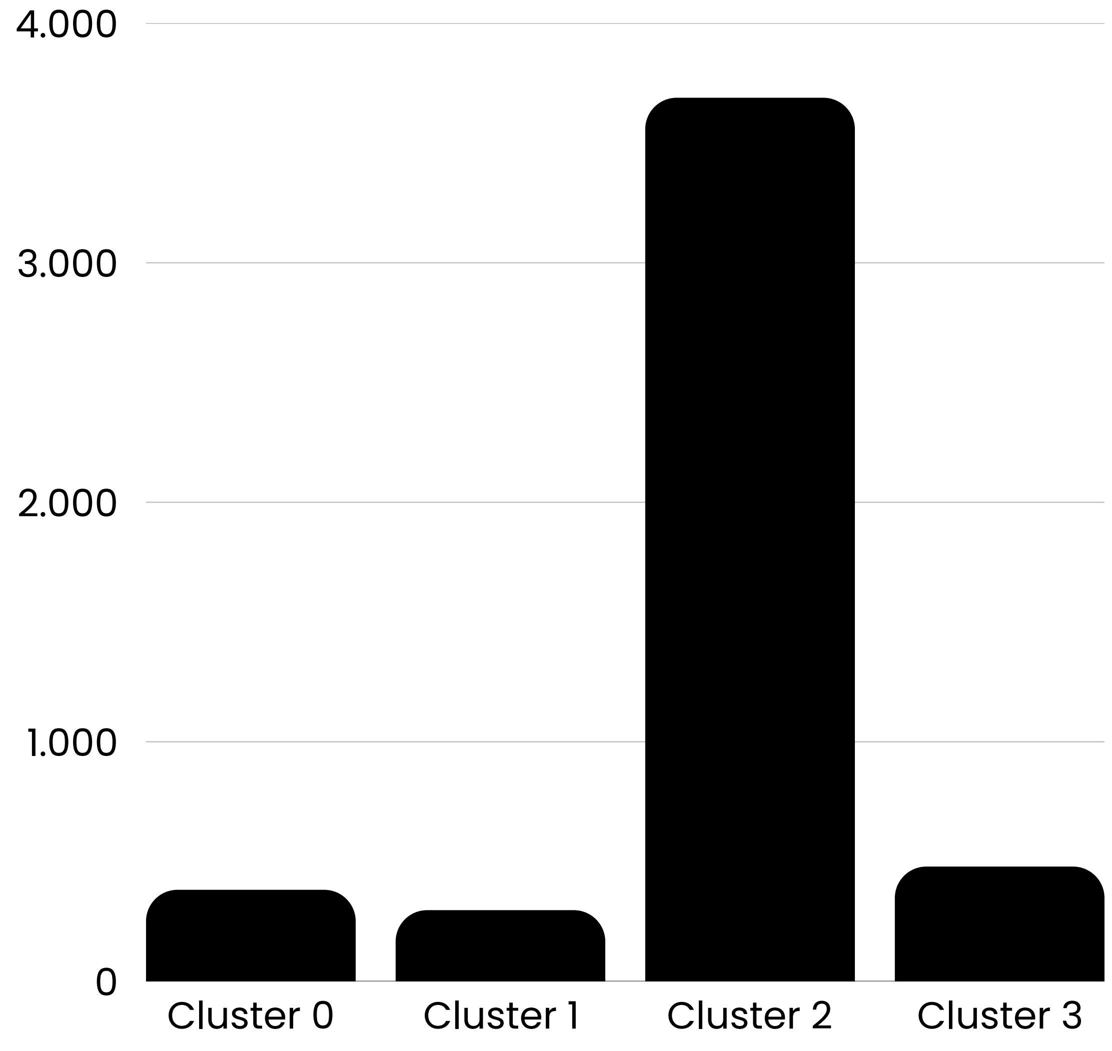
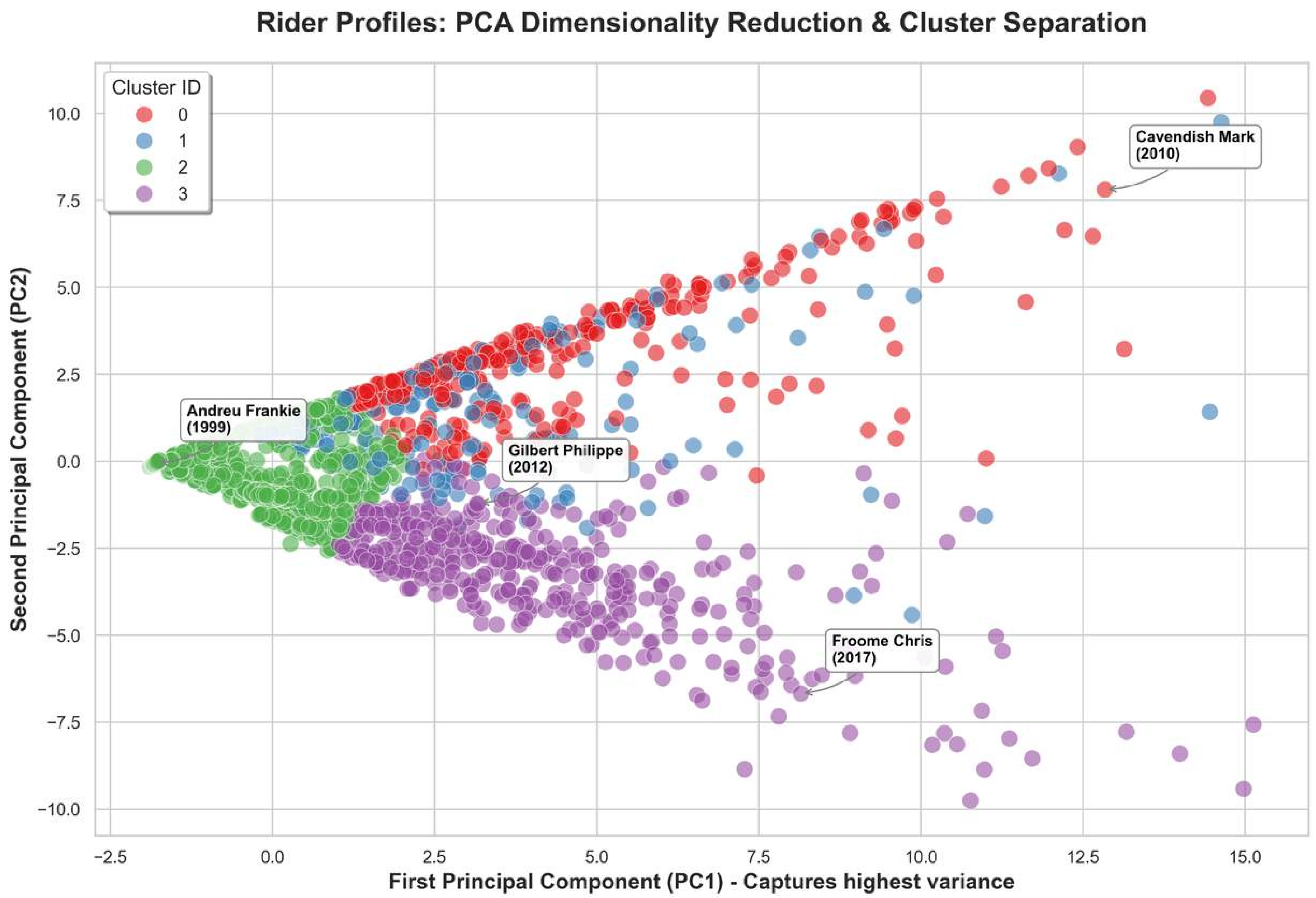




1 Analysis

CLUSTER SIZES REVEAL THE STRUCTURE OF THE PELOTON





1 Analysis

CLUSTER SIZES REVEAL THE STRUCTURE OF THE PELOTON

Cluster	Interpretation	Key evidence
0	Flat-stage sprinters	High flat points, flat top10s, stage wins
1	Hilly stage hunters	High hilly points and hilly top10s
2	Low stage-result visibility	Low points, few top10s, almost no wins
3	GC-oriented mountain/TT profiles	High mountain/TT points, high top10_gc

0
Sagan 2012

1
Sagan 2018

2
Bennati 2009

3
Nibali 2014

1 Analysis

EXTERNAL VALIDATION: Do clusters align with race outcomes?

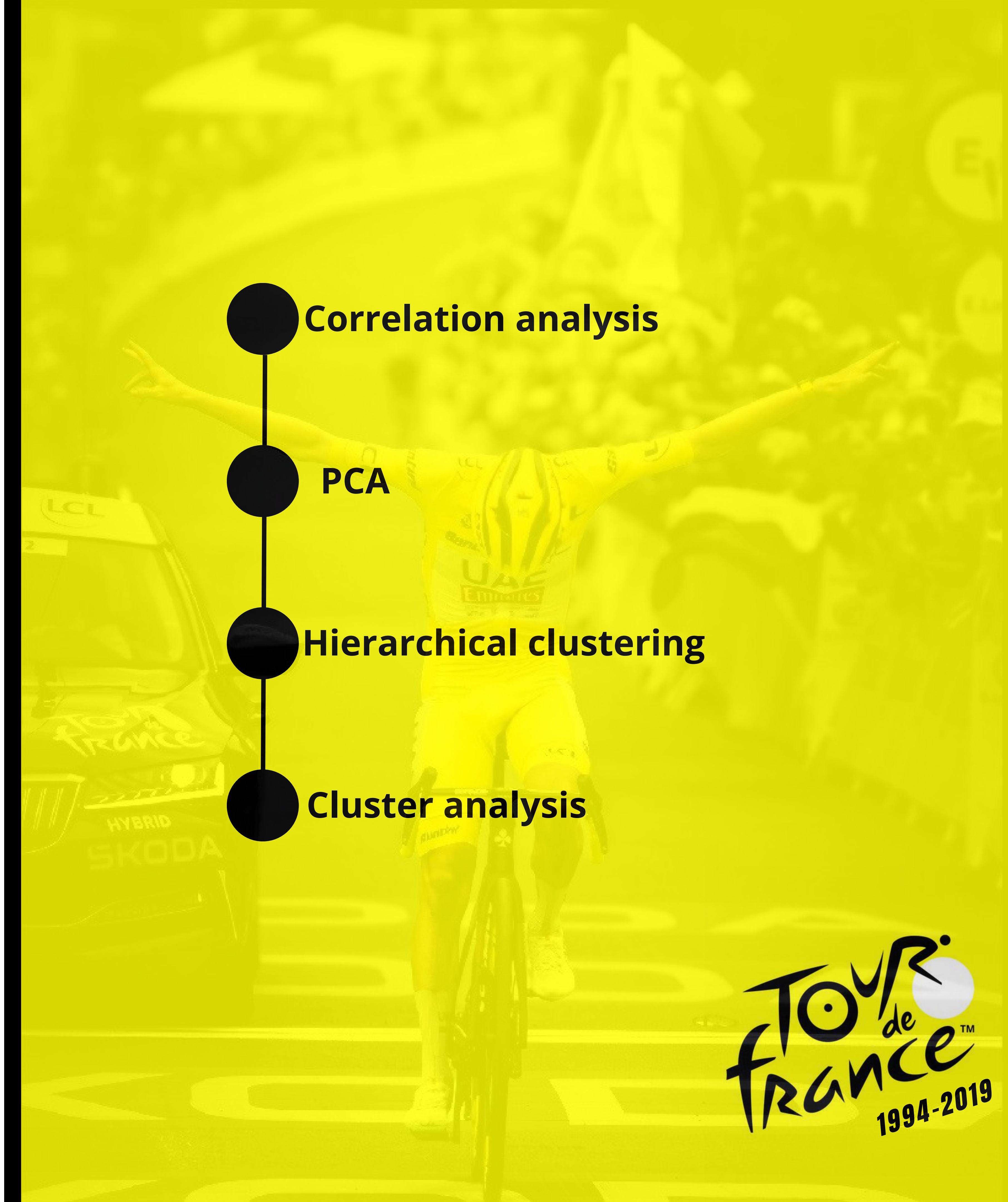
Cluster	final_rank	finished	top10_gc
0	100.5	73.8	0.008
1	54.2	87.2	0.057
2	82.2	73.2	0.005
3	23.7	90.2	0.428

2 Analysis

RIDER _ MERGED

OBJECTIVE:

The rider-merged analysis aims to identify **stable rider profiles** and long-term cycling specializations by **aggregating** each cyclist's Tour de France performance **into a single overall sporting identity.**



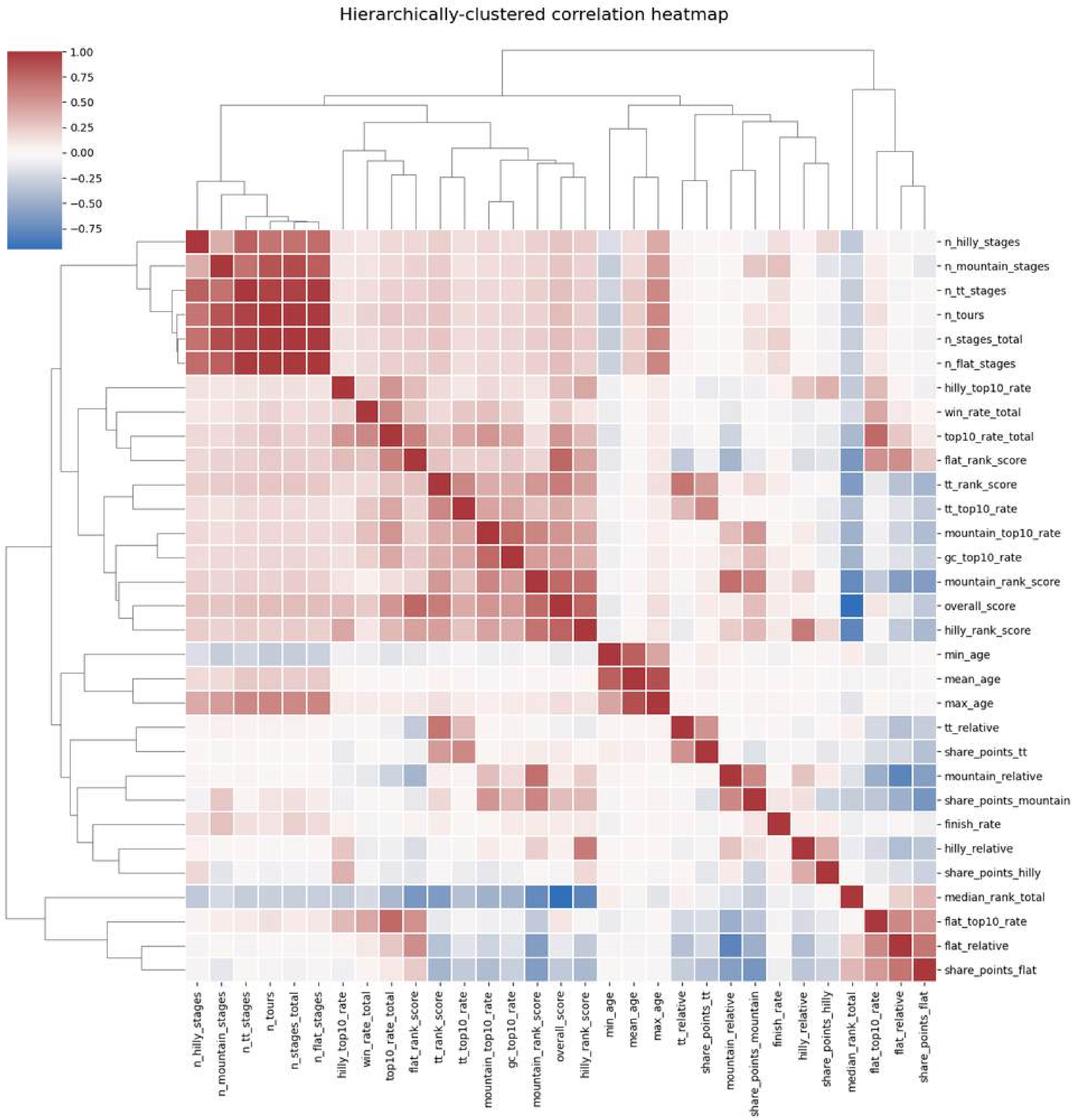
- Correlation analysis
- PCA
- Hierarchical clustering
- Cluster analysis

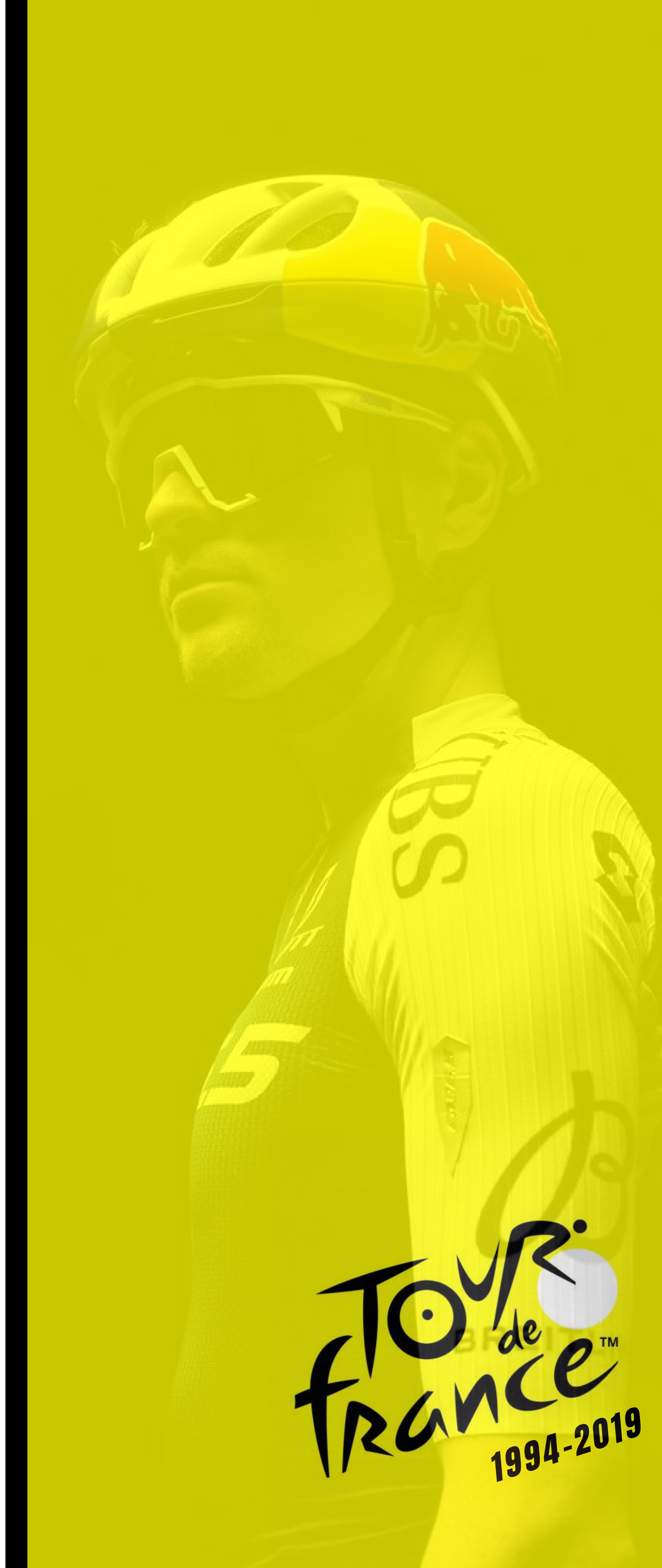
Tour de France 1994-2019

2 Analysis

CORRELATION ANALYSIS:

- VIF analysis
- correlation matrix
- Hierarchical cluster headmap

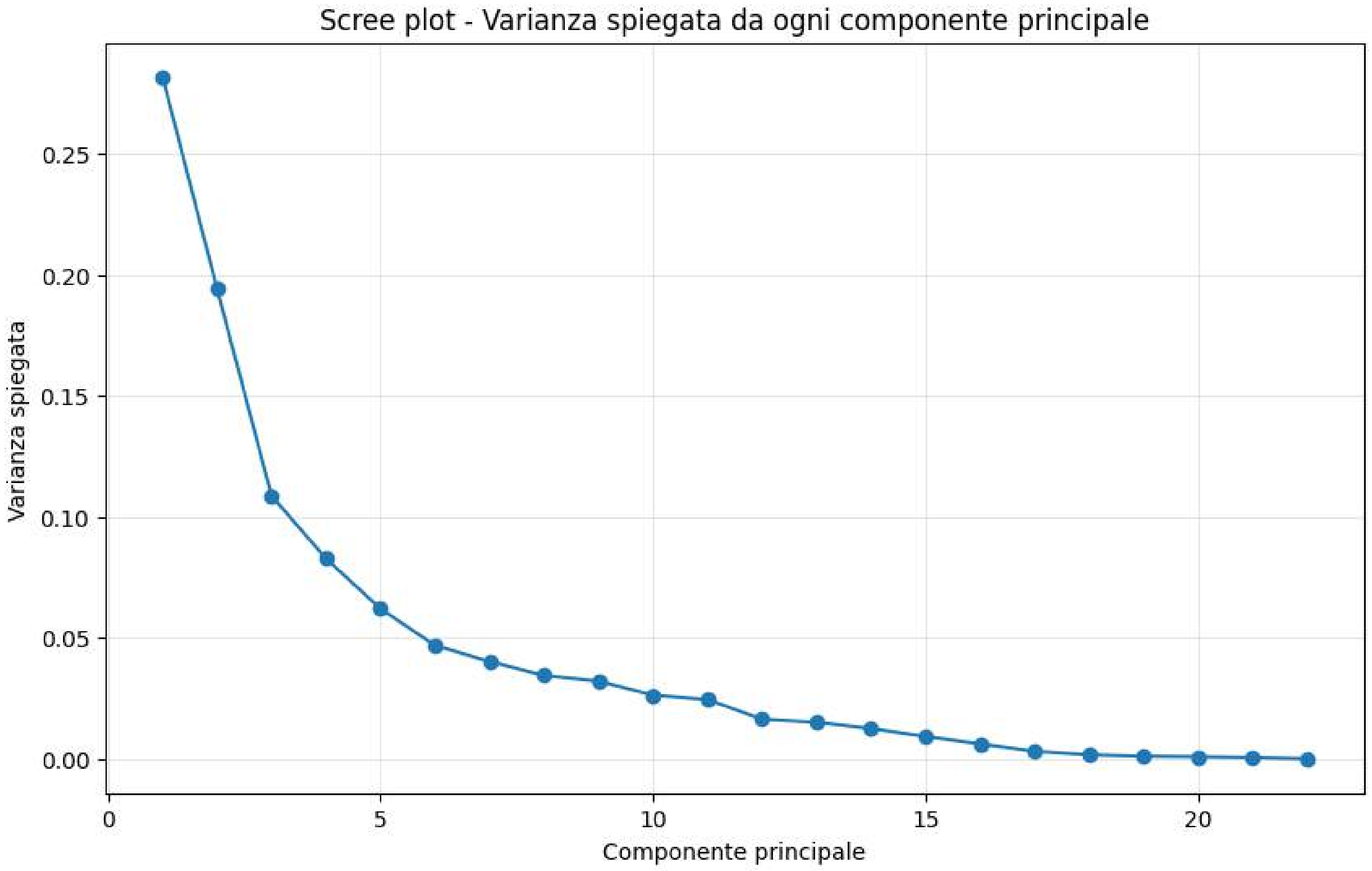


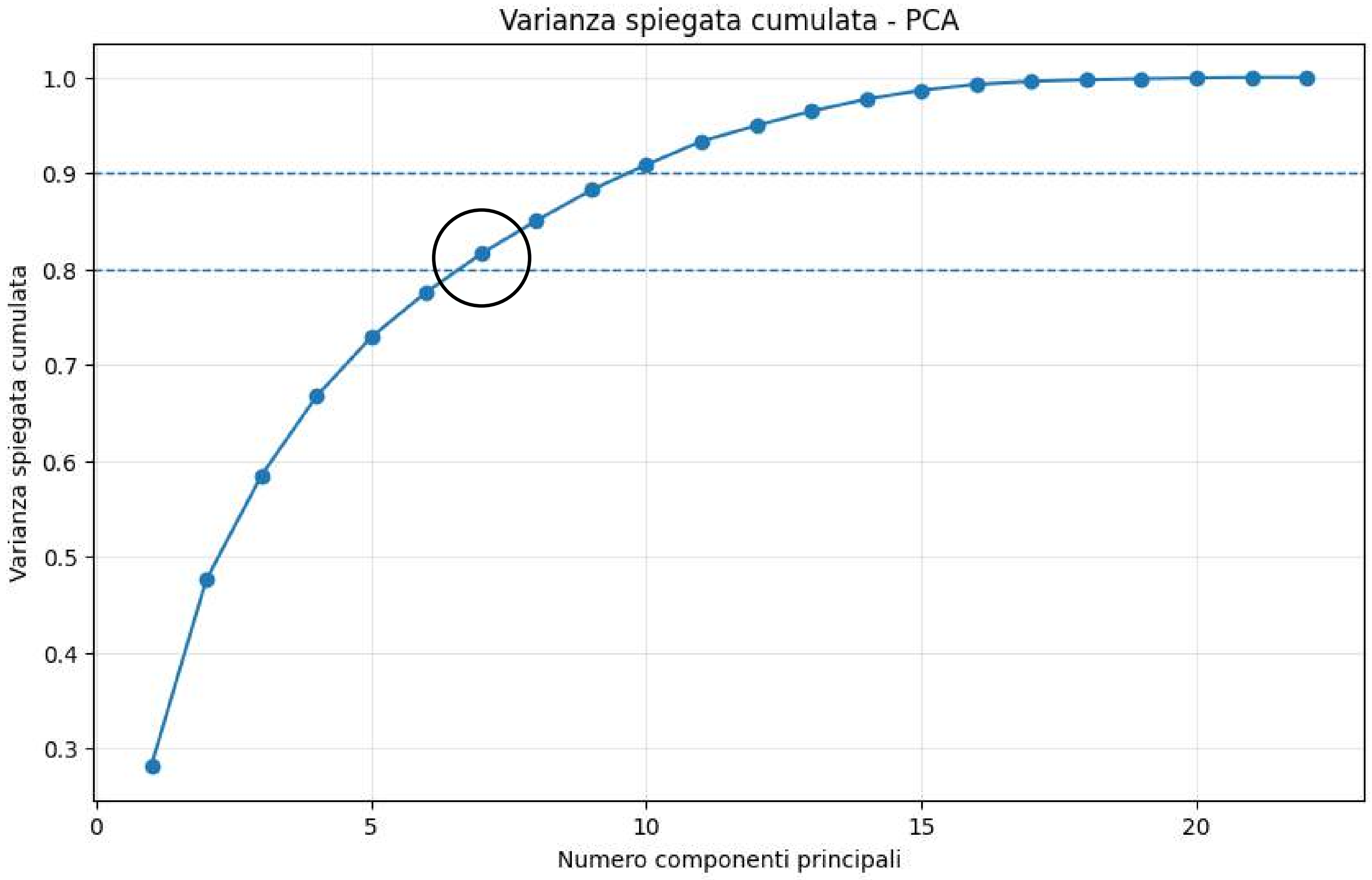


2 Analysis

PCA

- 7 principal components
- 81% of the total variance





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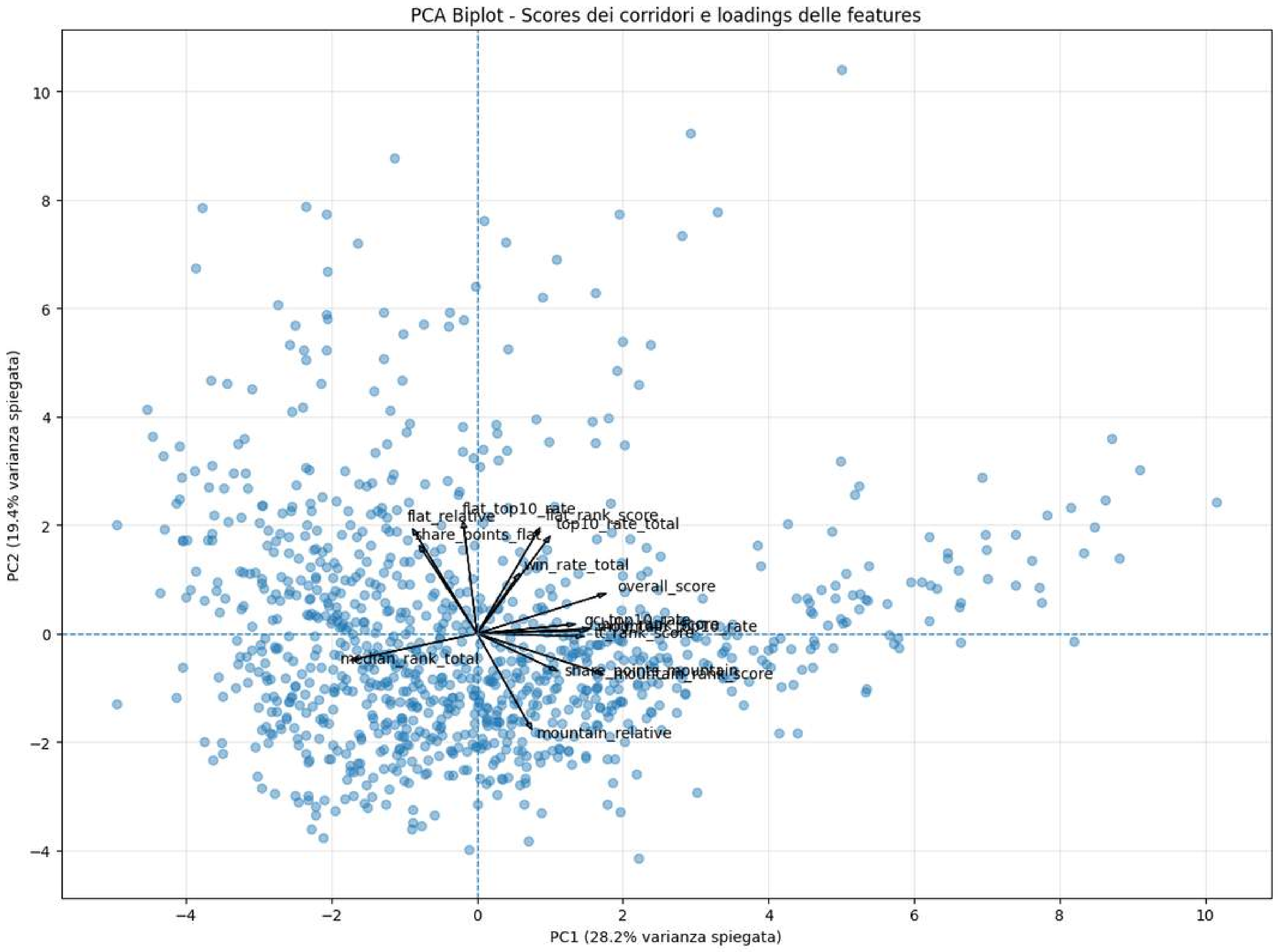
2 Analysis

1994-2019



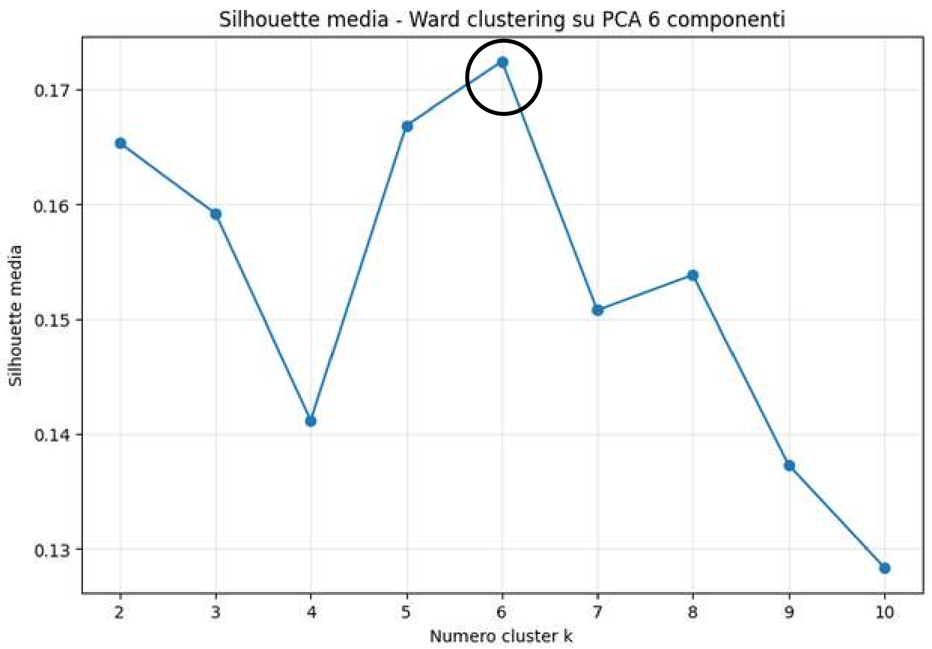
LOADINGS FOR PC1 AND PC2

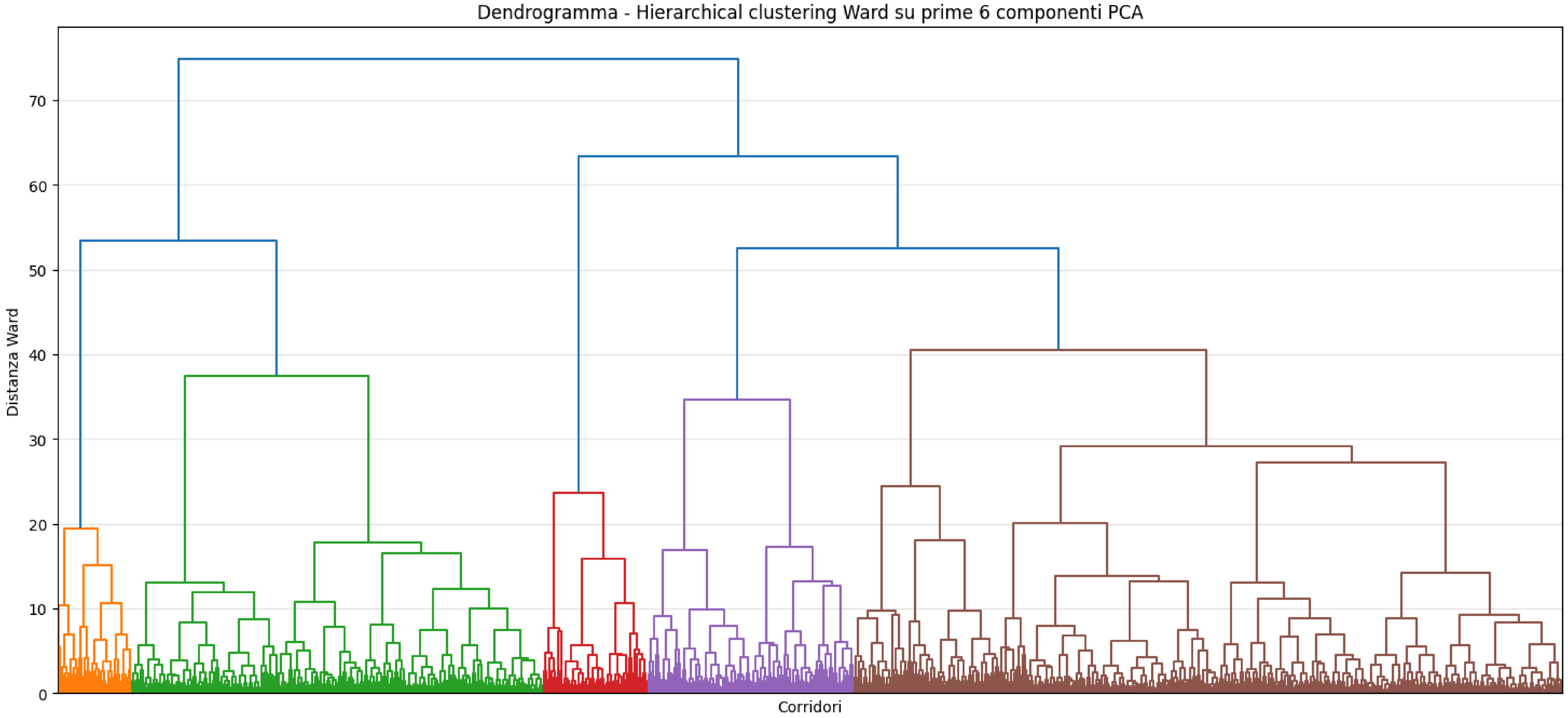
- **PC1** mainly represents **overall rider performance** and general competitiveness.
- **PC2** mainly captures the **contrast between flat/sprint-oriented** riders and **mountain-oriented** riders.

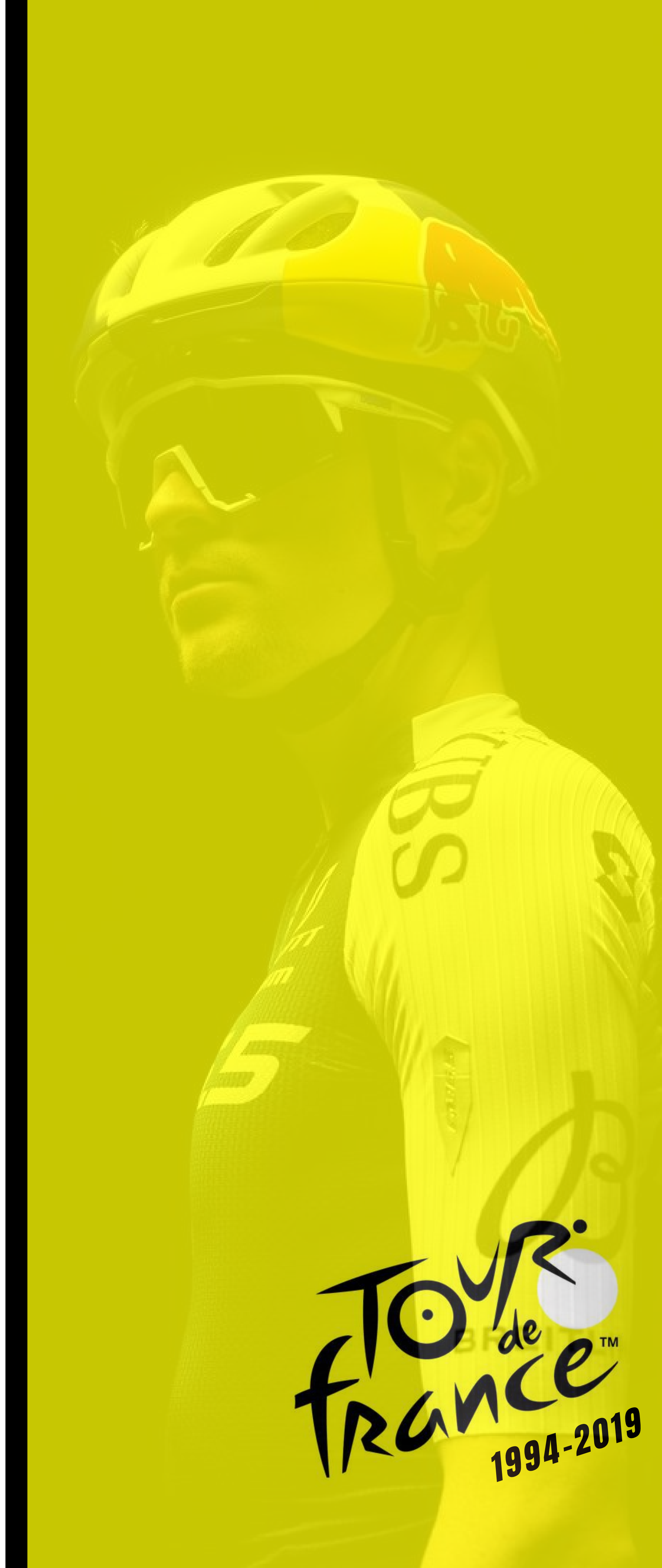
2 Analysis

HIERARCHICAL CLUSTERING

- Clusters based on **PCA components**
- Using **Ward linkage**
- Cut at **k = 6.**

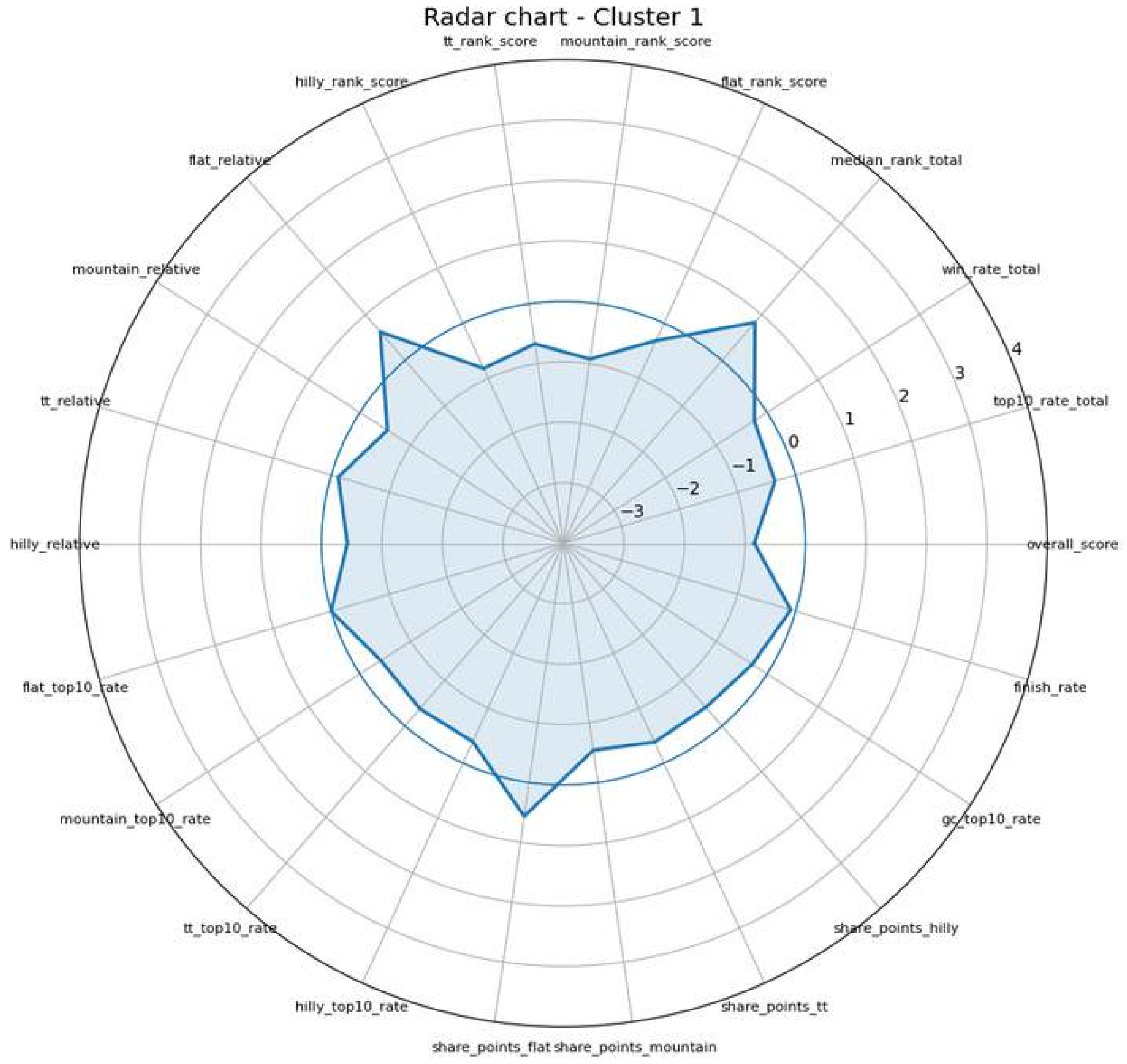


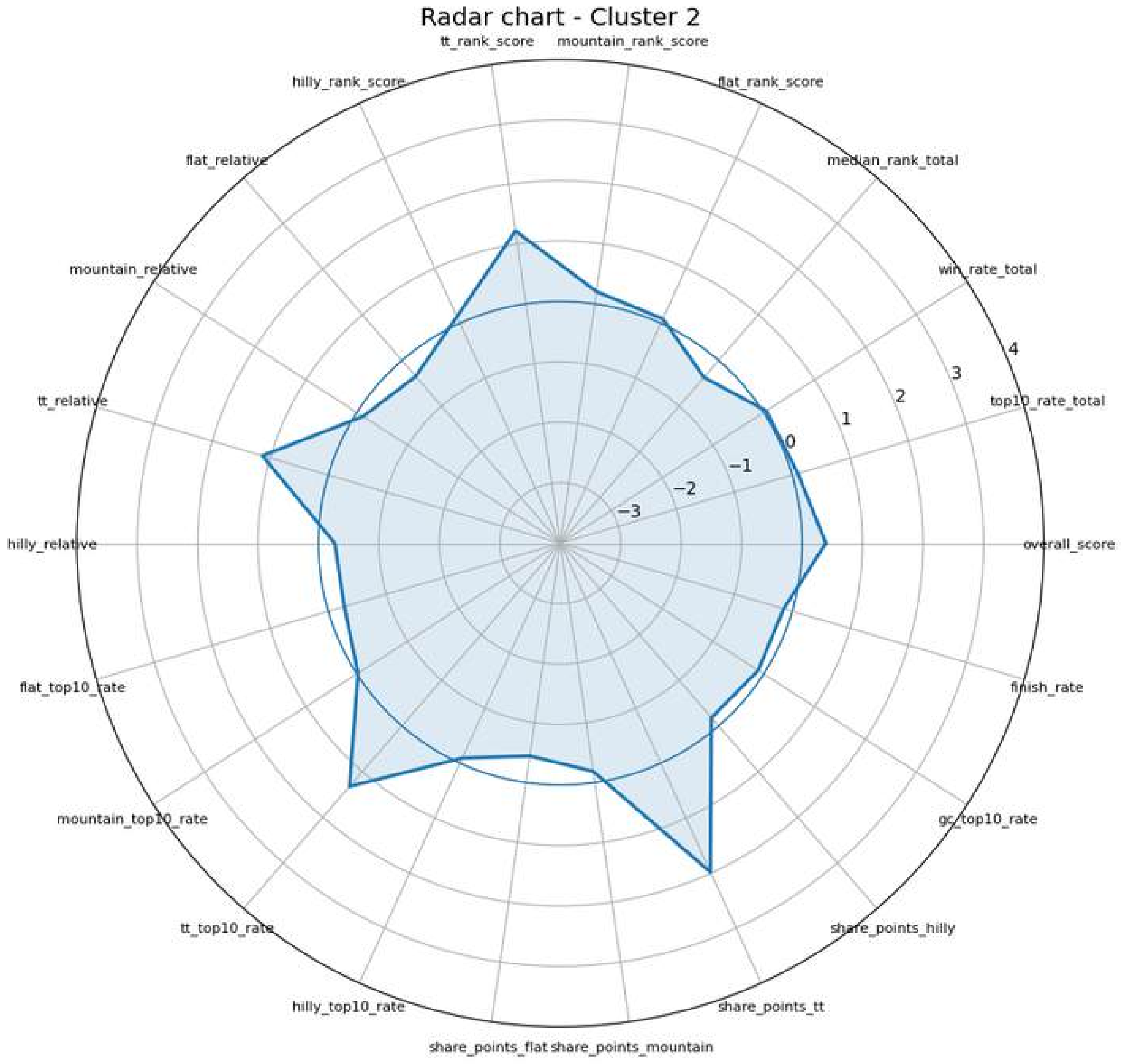


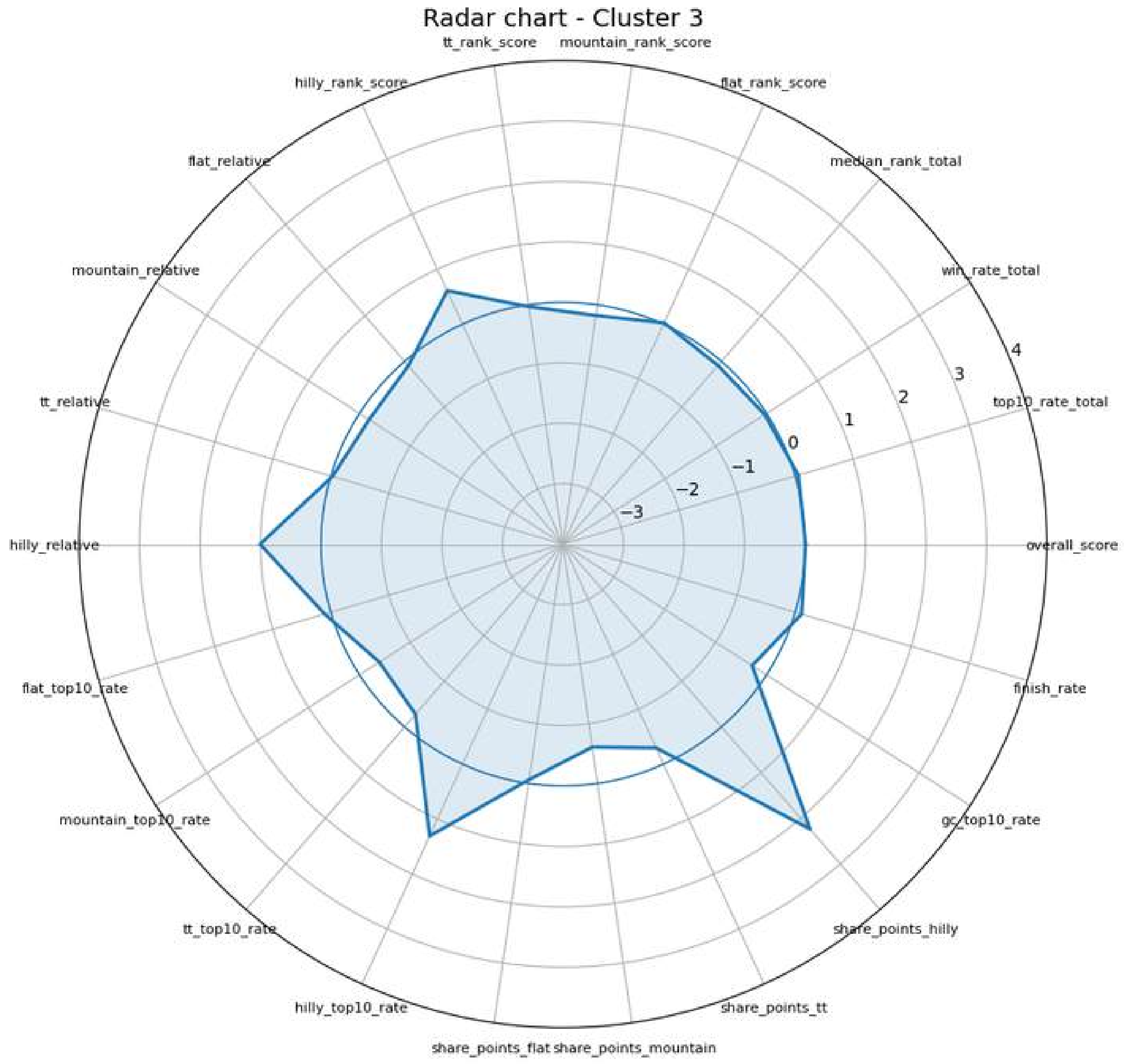


CLUSTER INTERPRETATION

Flat-oriented	Crono-men	Hilly-puncheurs



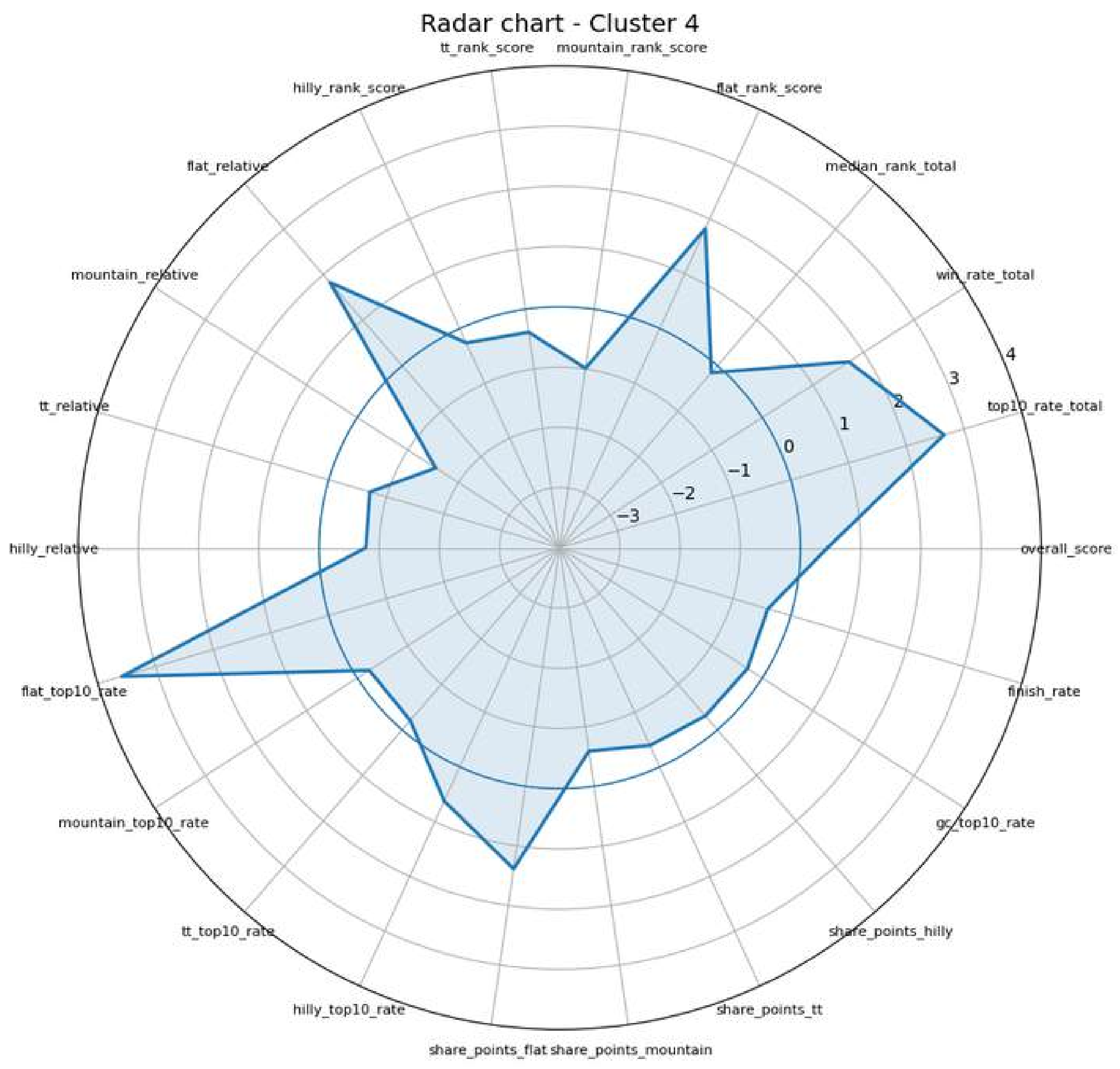




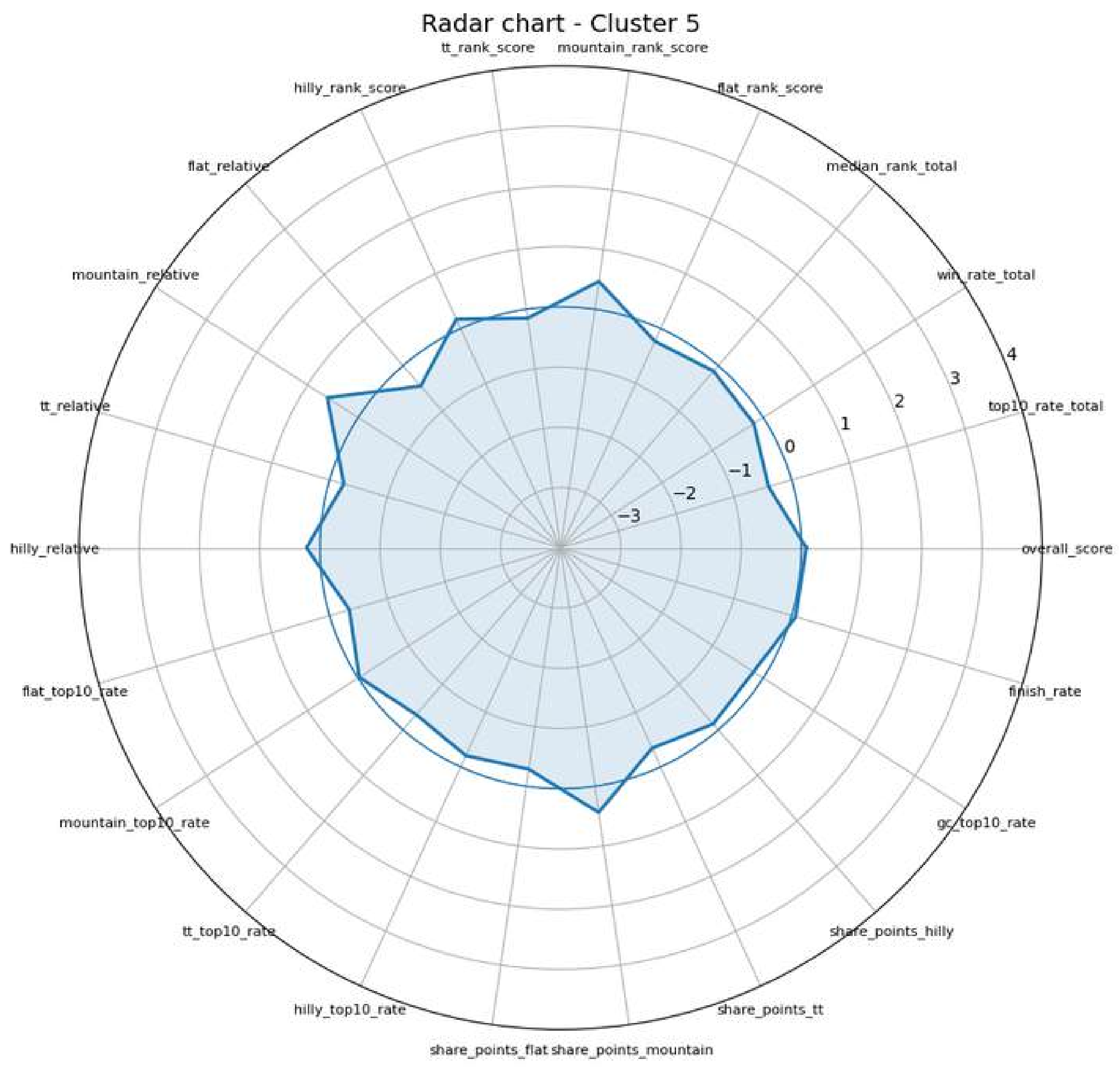
CLUSTER INTERPRETATION

1994-2019

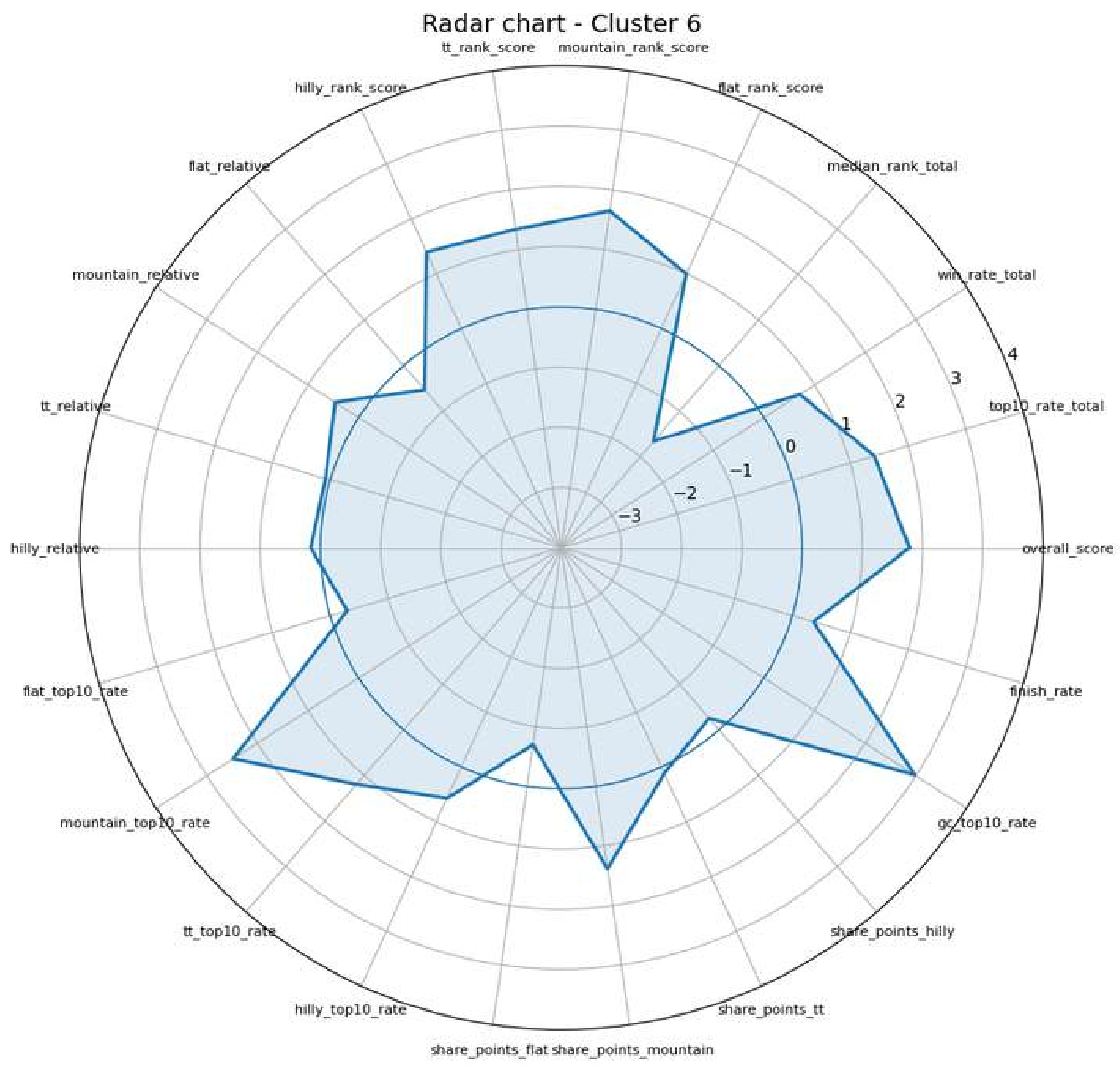
Sprinters



Mountain-oriented

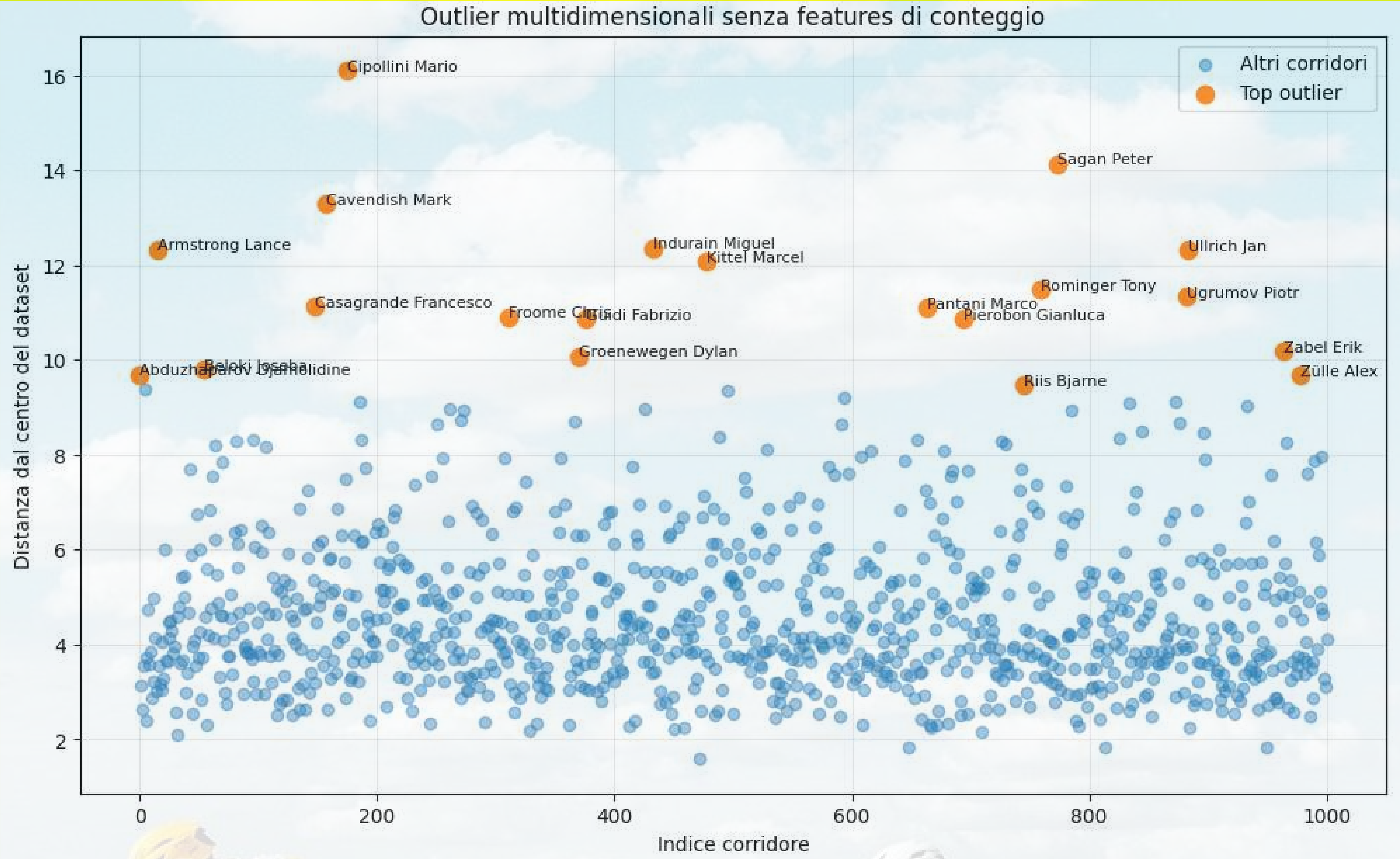


GC men



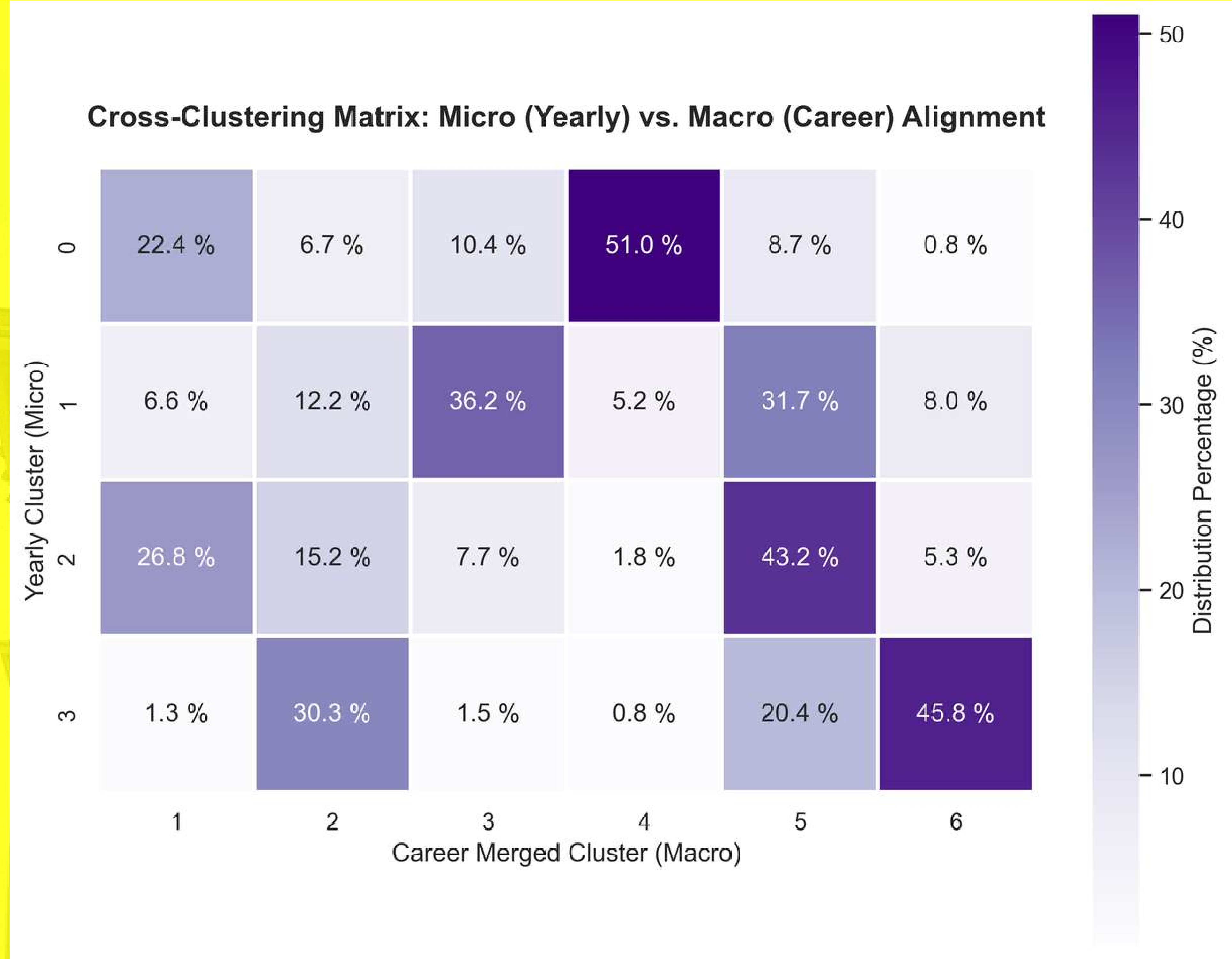
OUTLIERS HANDLING:

- With and without outlier removal, the **main clusters remain identifiable**, which supports the **robustness of the analysis**
- Without outliers, the resulting clusters become **more moderate** and **less dominated** by extreme specialist profiles.
- Outliers were not necessarily errors; instead, they could represent **meaningful sporting signals.**



CROSS-CLUSTERING ANALYSIS

- The core objective of this analysis is to **validate the statistical consistency** between year-to-year and career performances
- The **Cross-Clustering Matrix**: Rows represent the cluster assigned to a specific rider in a single year, normalized to show where those performances "flow" across final career categories (columns)
- Key findings: high correspondence for pure sprinters and GC/climbers, more dispersed data in the “not so pure” categories



Tour de France 1994-2019

CASE STUDY: VALIDATING CLUSTERS THROUGH CYCLING LEGENDS

Tour de France 1994-2019

- **Longitudinal Analysis** (1994–2019): Cross-referencing micro (yearly) and macro (career) clustering results for four iconic riders.
- **Strategic Selection**: Lance Armstrong, Sylvain Chavanel, Peter Sagan, and Philippe Gilbert were chosen to represent completely different tactical profiles.
- The Ultimate Check: Evaluating **how the unsupervised algorithms capture specialization,** while highlighting model limitations.

	CAREER_CLUS	1994	1995	1996	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Armstrong Lance	6	2.0	0.0	2.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0	-	-	-	2.0	3.0	-	-	-	-	-	-	-	-	-
Chavanel Sylvain	2	-	-	-	-	-	2.0	2.0	3.0	2.0	1.0	2.0	2.0	0.0	2.0	0.0	2.0	3.0	2.0	3.0	2.0	2.0	2.0	2.0	-
Gilbert Philippe	5	-	-	-	-	-	-	-	-	-	1.0	2.0	2.0	2.0	-	-	0.0	3.0	2.0	-	-	-	2.0	1.0	-
Sagan Peter	4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.0	0.0	0.0	0.0	0.0	-	1.0	1.0

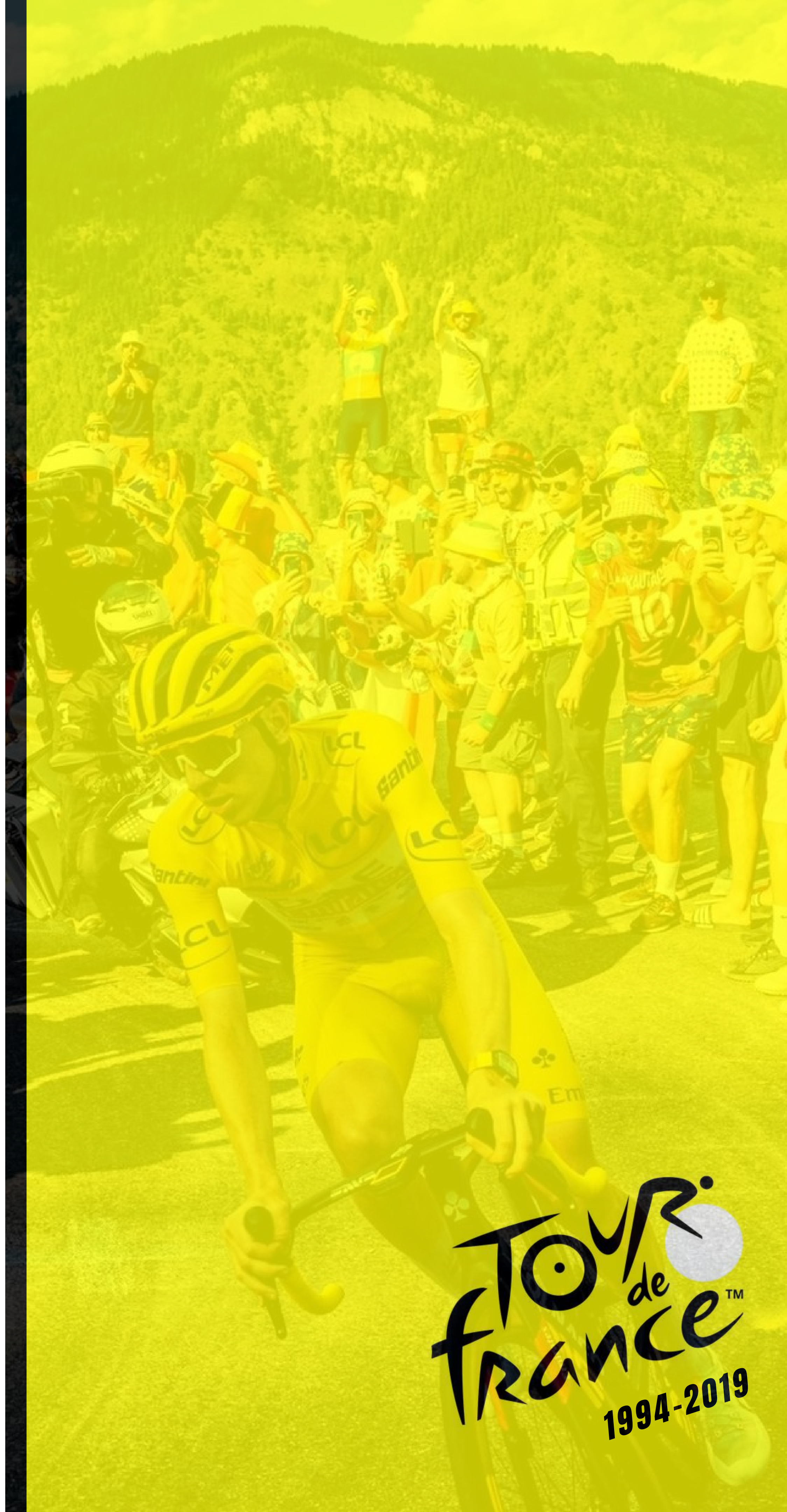
Conclusions: Do riders specialize? **Yes, but...**

Limits:

- Data incompleteness,
- Exclusive use of Tour de France data,
- Stage results bias (results as the only possible variable to assess general performances),
- The *"Anonymous Rider" problem* in year-to-year clustering method

Future research questions:

- Richer datasets such as *procyclingstats.com*,
- Expand racing horizons (other Grand Tours, but also one-day Classics),
- Transition to supervised learning: a top 10 prediction
- Compare specialization throughout different time periods
- Expand the clusterization in other sports



THANK YOU FOR THE ATTENTION

And


VIVE
LE
TOUR

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